FAKULTA APLIKOVANÝCH VĚD
ZÁPADOČESKÉ UNIVERZITY
V PLZNI

Edgebreaker

Komprese trojúhelníkových sítí

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2026-11-03

1. Motivace

1.1 Dobový kontext

1.2 Dobový kontext

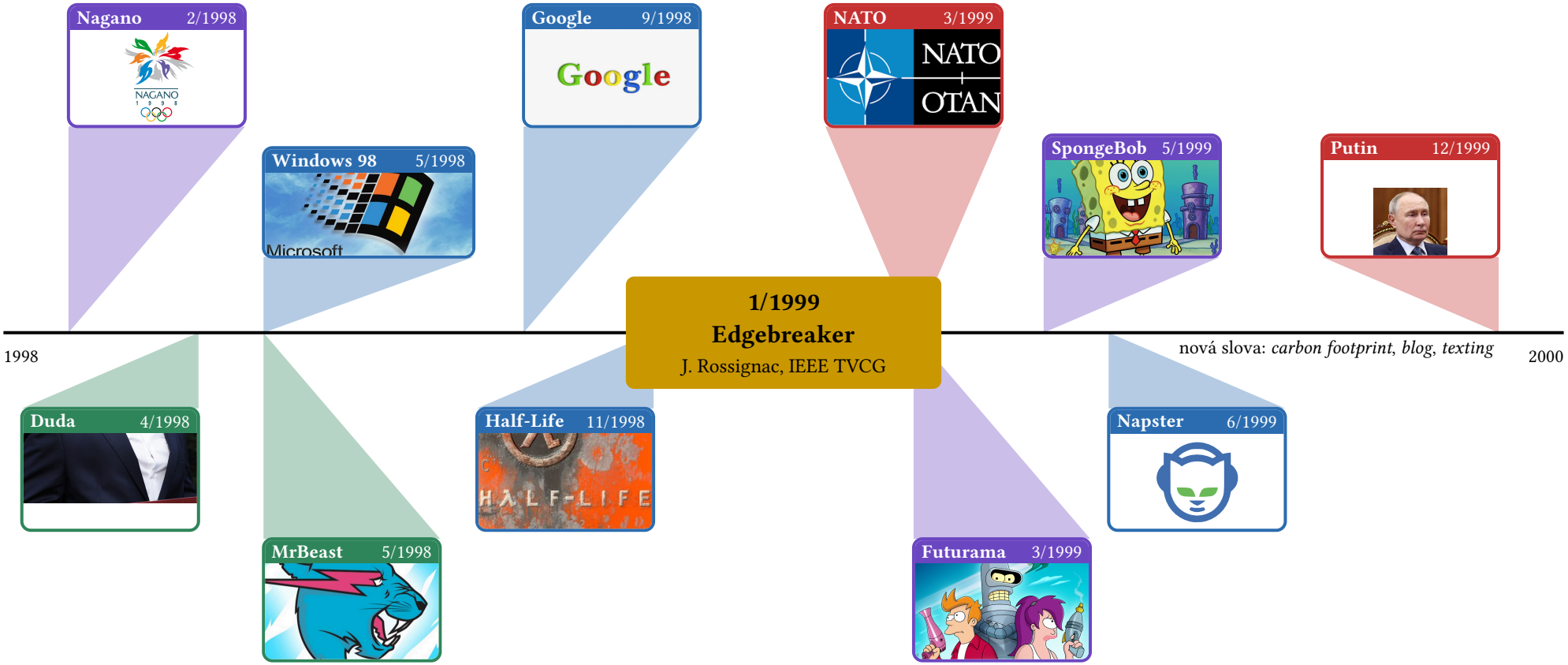


Figure 1: Dobový kontext vzniku algoritmu Edgebreaker.

1.2 Dobový kontext

nagano: By [1], Fair use, https://en.wikipedia.org/w/index.php?curid=14328465 win98: https://i1.sndcdn.com/artworks-TFheyNniW28o9wre-ddA99Q-t1080x1080.jpg google: https://logodix.com/logo/55299.png nato: https://www.nato.int/content/experience-fragments/nato/site/header/english/master/_jcr_content/root/c06_main_header_copy/logo.coreimg.png/1789636393643/nato.png sponge bob: https://media.craiyon.com/2025-04-21/FWs1FwNsS3WCDomSej0k8w.webp putin: Autor: Kremlin.ru, CC BY 4.0, https://commons.wikimedia.org/w/index.php?curid=163363642 duda: Autor: Gov.pl, CC BY 3.0 pl, https://commons.wikimedia.org/w/index.php?curid=128272581 mrbeast: https://yt3.ggpht.com/nxYrc_1_2f77DoBadyxMTmv7ZpRZapHR5jbuYe7PlPd5cIRJxtNNEYyOC0ZsxaDyJJzXrnJiuDE=s800-c-k-c0x00ffffff-no-rj half-

life: By May be found at the following website: https://valvearchive.com/archive/Half-Life/Half-Life/Art/Box%20Art/front2.jpg, Fair use, https://en.wikipedia.org/w/index.php?curid=13514455 futurama: (csfd) https://image.pmgstatic.com/cache/resized/w663/files/images/film/posters/168/087/168087719_geyysj.png napster: Autor: Jusore – Vlastní dílo, GPLv3, https://commons.wikimedia.org/w/index.php?curid=169974746

1.3 Dělení

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1.3.1 Geometrie

$$\forall v \in V :$$

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$$\forall v \in V :$$

$$v = (x, y, z)$$

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$$\forall v \in V :$$

$$v = (x, y, z, n, \mathrm{uv}, \ldots)$$

1.3.2 Konektivita

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$$\forall v \in V :$$

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1.3.2 Konektivita

$$\forall e \in E : e = \left(e_i, e_j\right)$$

1.3 Dělení

1.3.1 Geometrie

$$\forall v \in V :$$

$$v = (x, y, z, n, \mathrm{uv}, \ldots)$$

1.3.2 Konektivita

$$\forall e \in E : e = \left(e_i, e_j\right)$$

$$\forall t \in T : t = \left(v_i, v_j, v_k\right)$$

1.3 Dělení

1.3.1 Geometrie

$$\forall v \in V :$$

$$v = (x, y, z, n, \mathrm{uv}, \ldots)$$

1.3.2 Konektivita

$$\forall e \in E : e = \left(e_i, e_j\right)$$

$$\forall t \in T : t = \left(v_i, v_j, v_k\right)$$

$$\forall c \in 3T : c = \left(c_{\mathrm{op}}\right)$$

1.3 Dělení

1.3.1 Geometrie

$$\forall v \in V :$$

$$v = (x, y, z, n, \mathrm{uv}, \ldots)$$

1.3.2 Konektivita

$$V : 2F : 3E$$

2. Komprese

2.1 Pravidla

History =

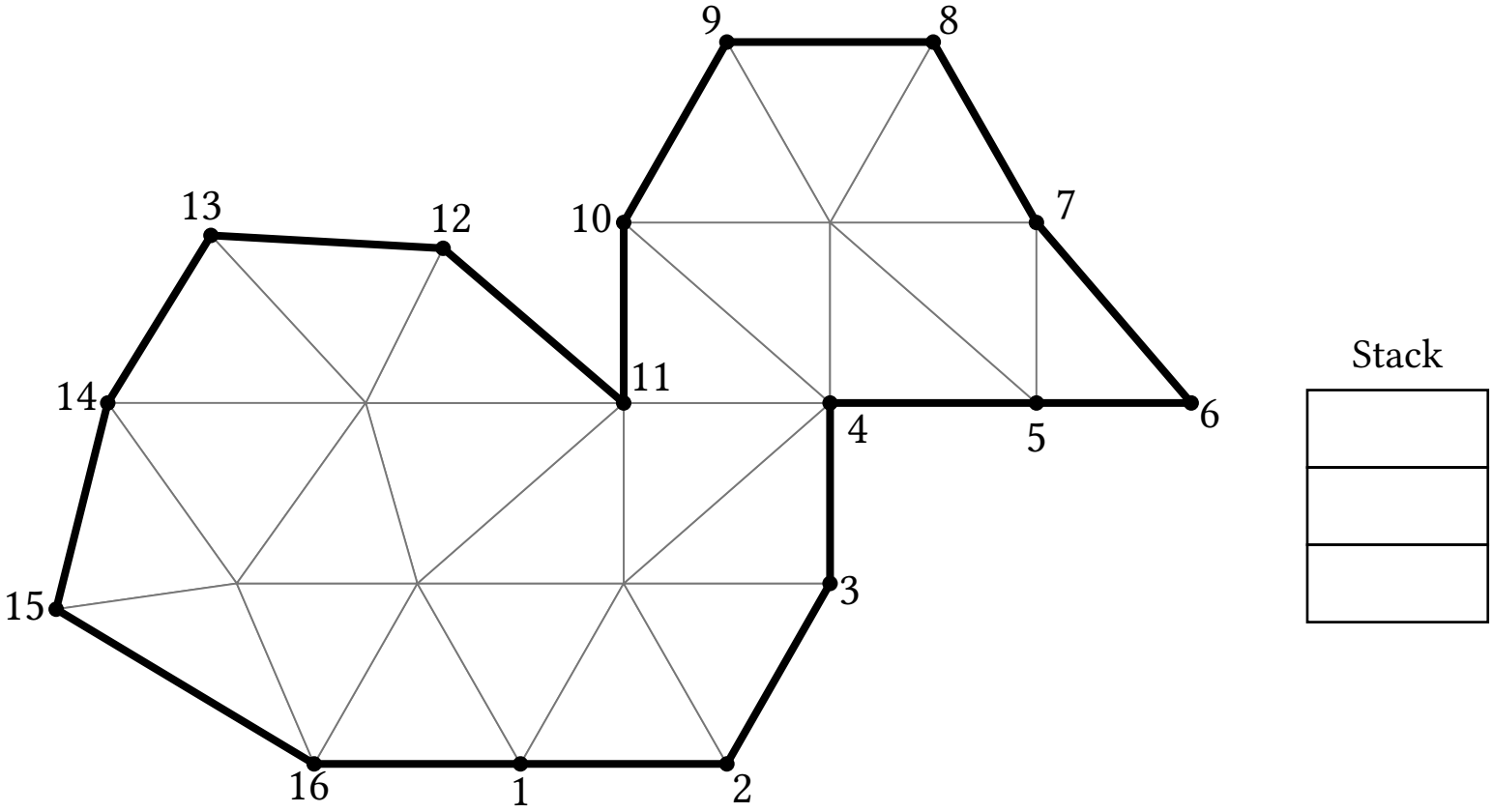


Figure 2: Jednoduchá komprese.

History = C

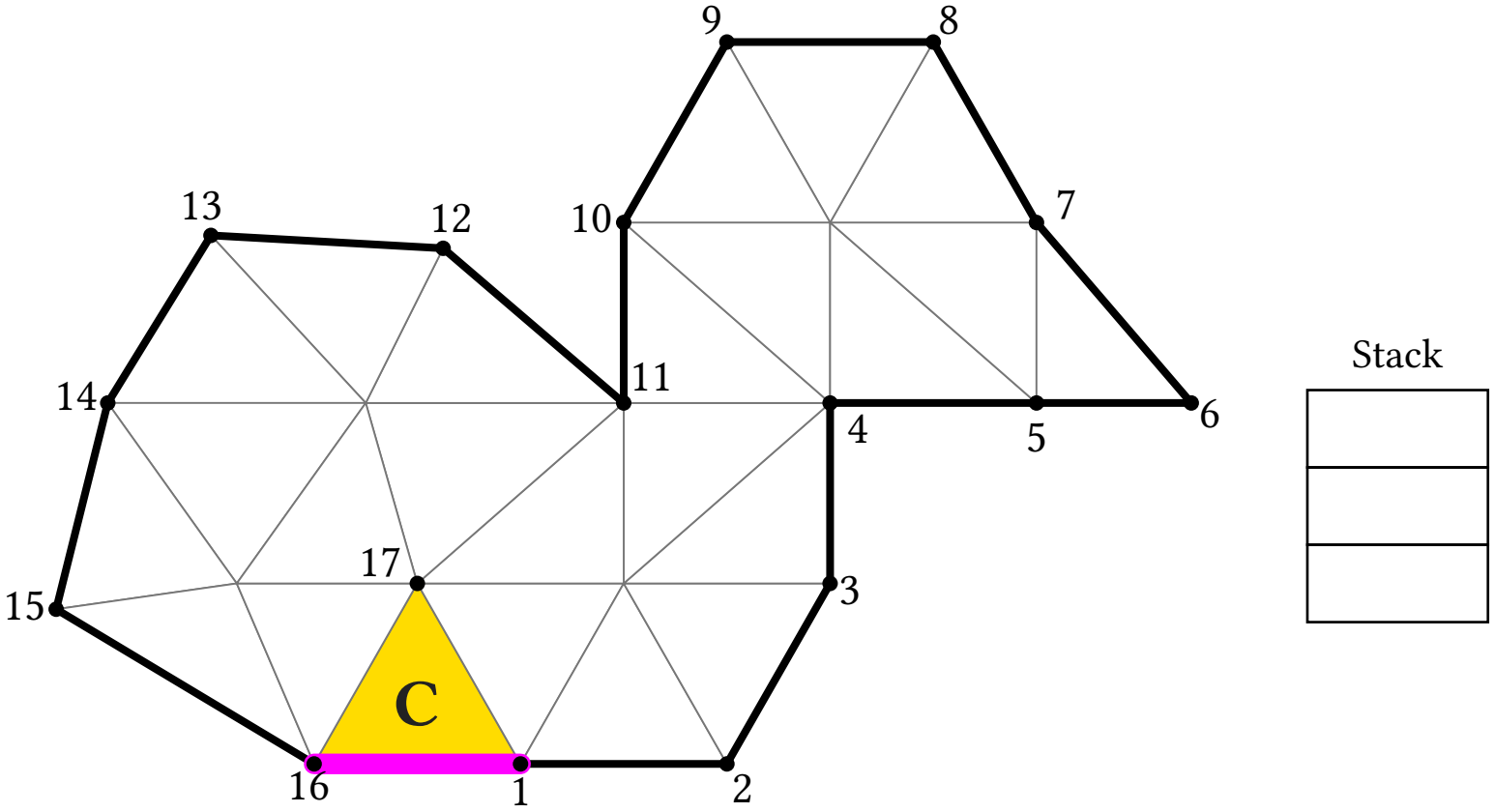


Figure 2: Jednoduchá komprese.

History = CC

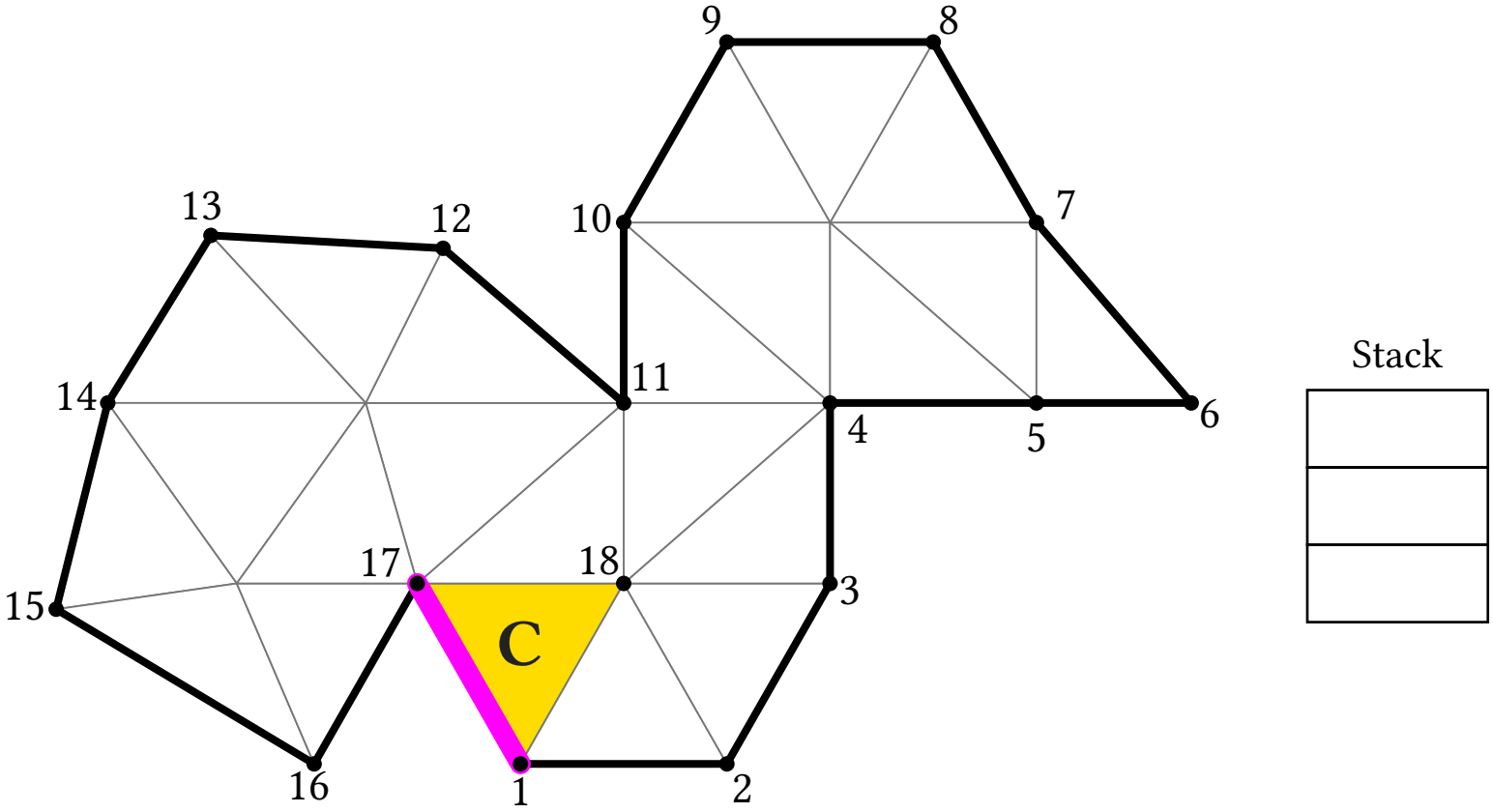


Figure 2: Jednoduchá komprese.

History = CCR

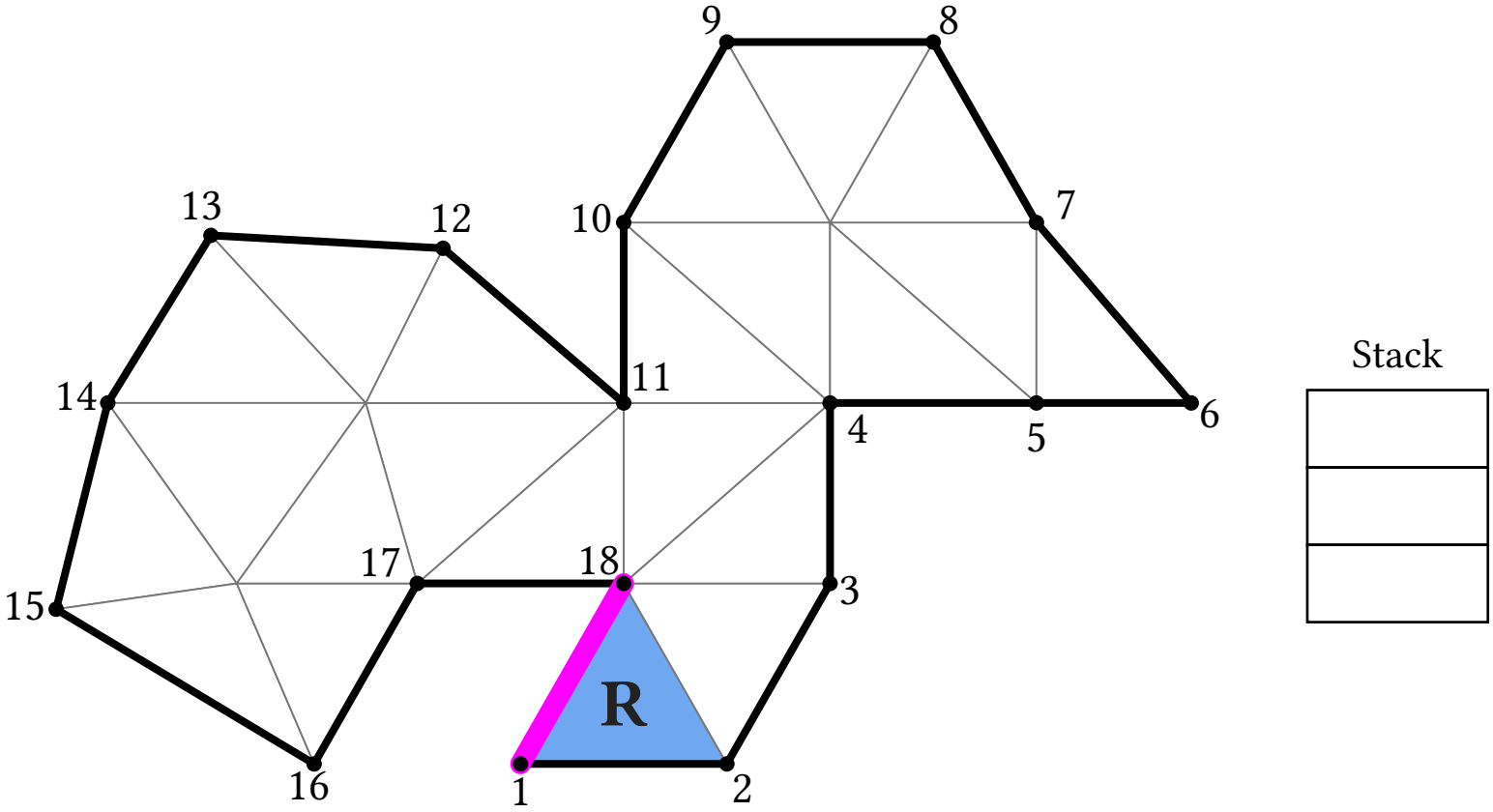


Figure 2: Jednoduchá komprese.

History = CCRR

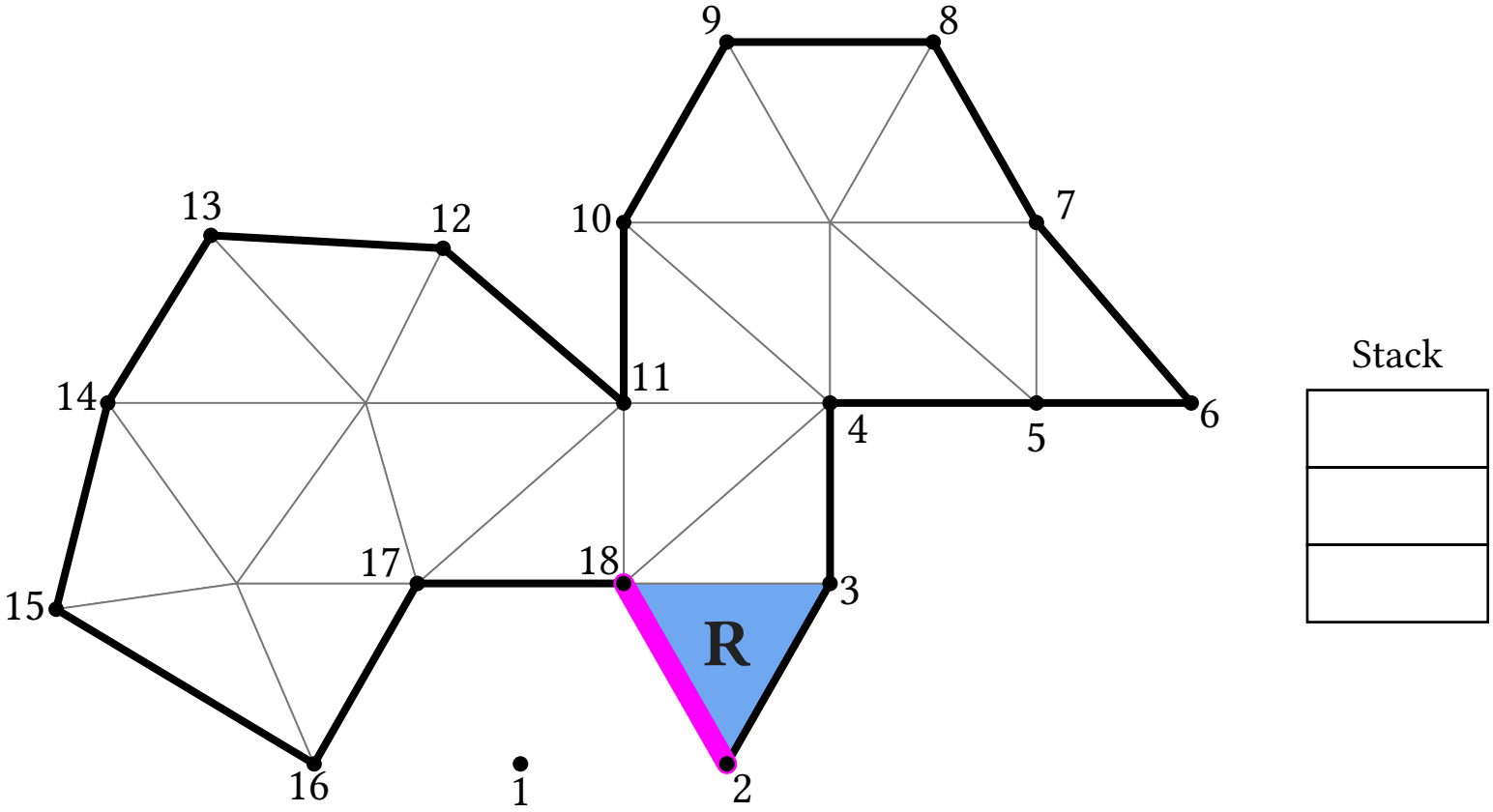


Figure 2: Jednoduchá komprese.

History = CCRRR

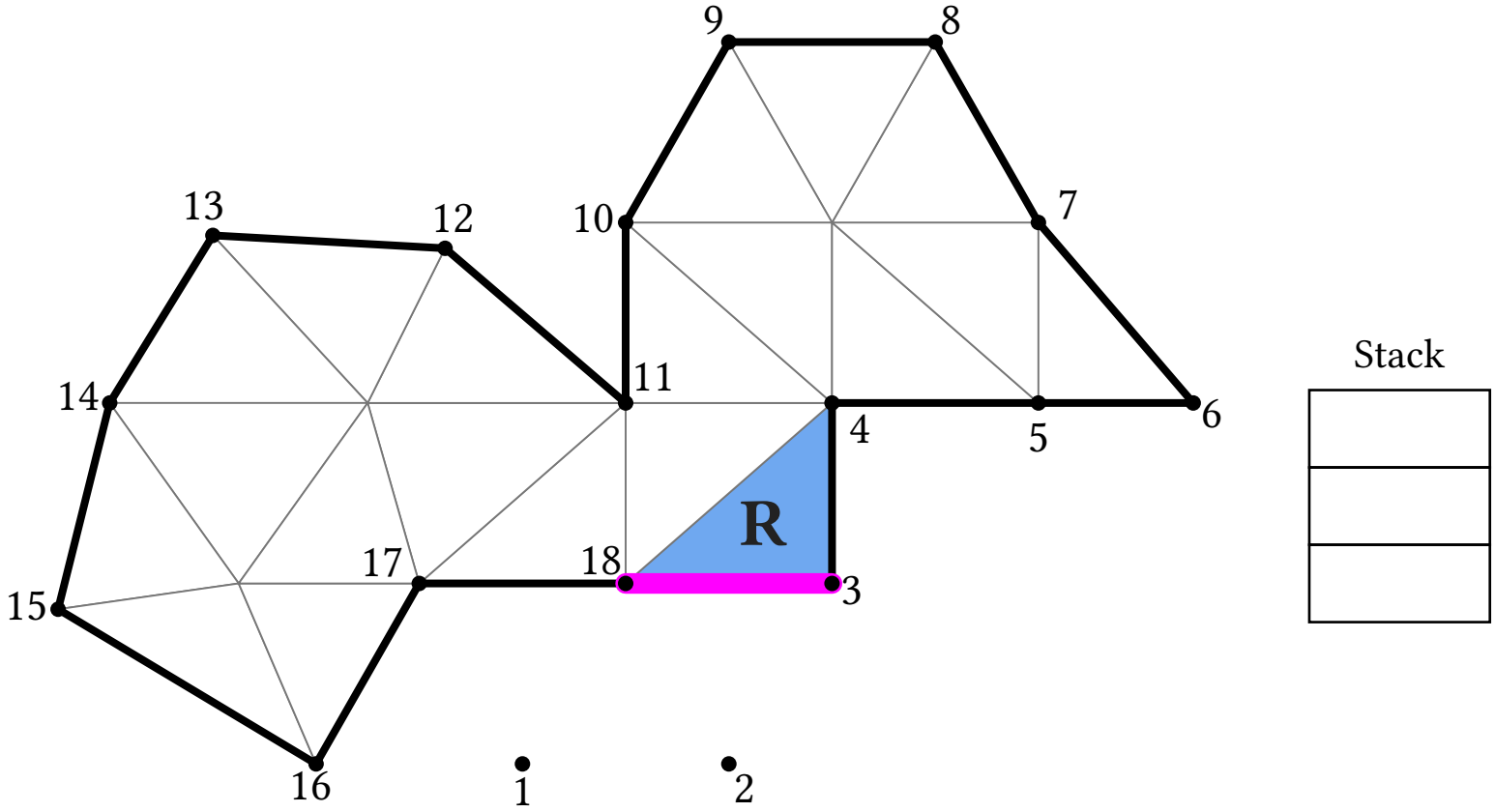


Figure 2: Jednoduchá komprese.

History = CCRRRS

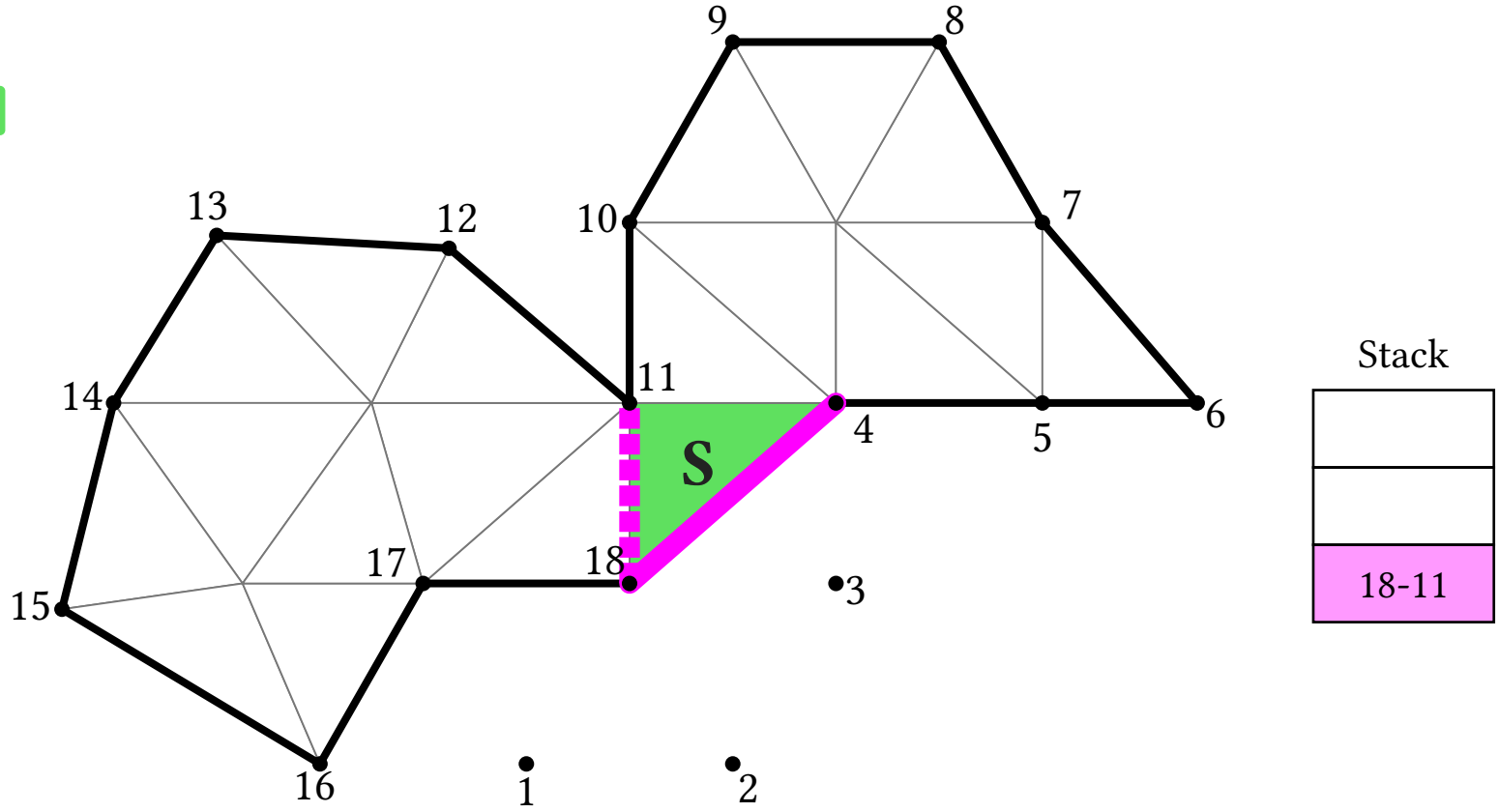


Figure 2: Jednoduchá komprese.

History = CCRRRSL

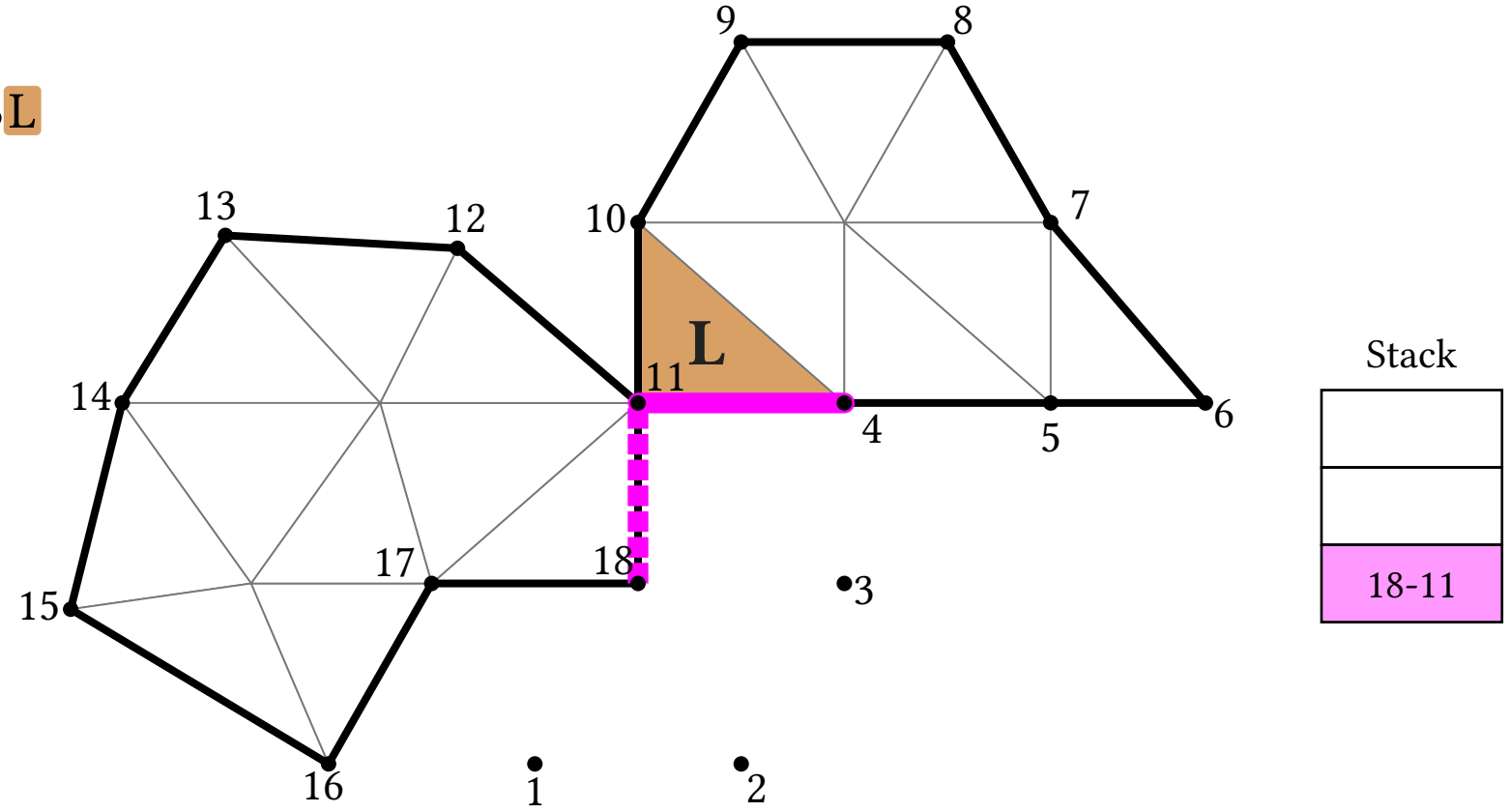


Figure 2: Jednoduchá komprese.

History = CCRRRSLC

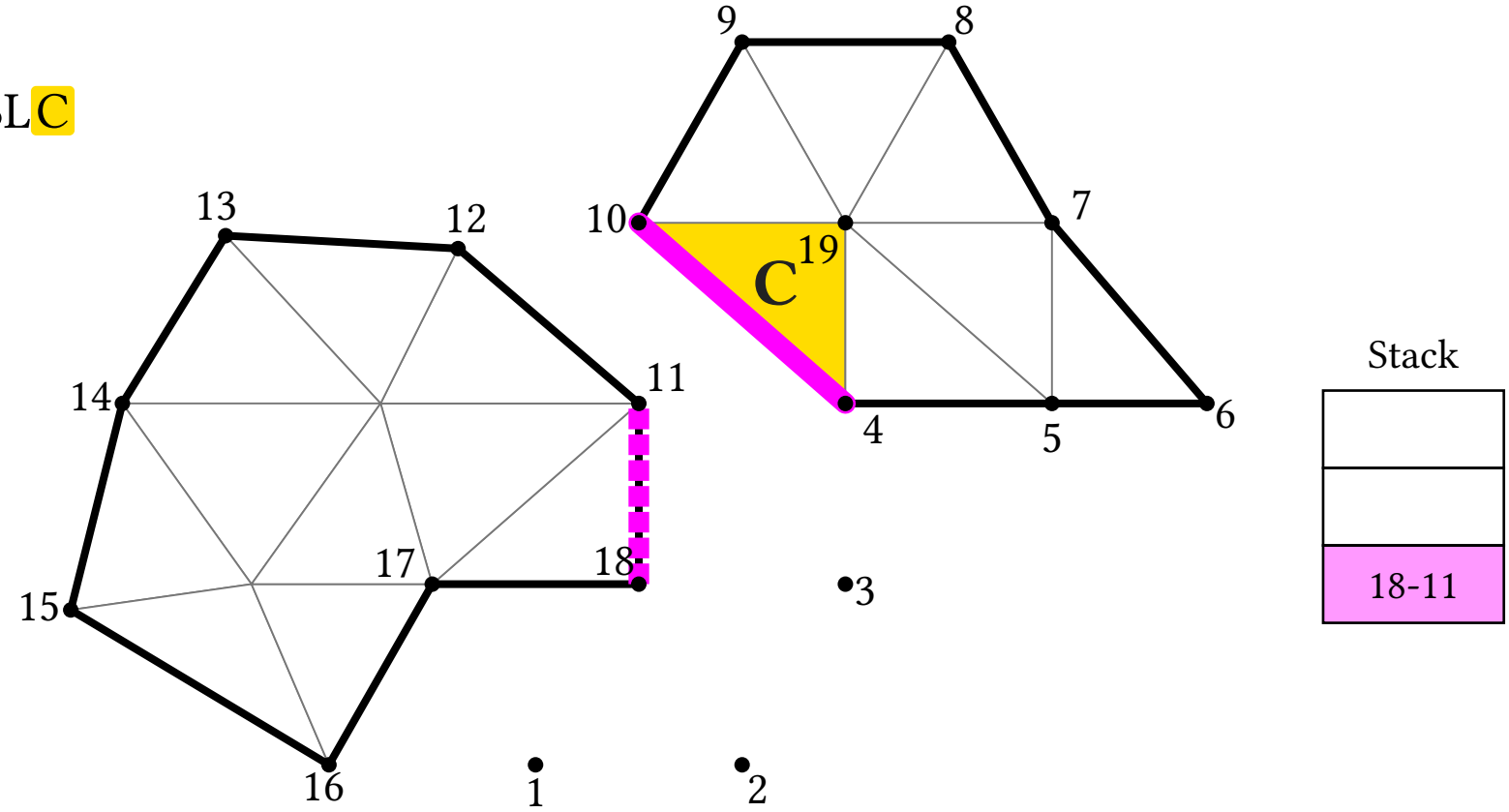


Figure 2: Jednoduchá komprese.

History = CCRRRSLCR

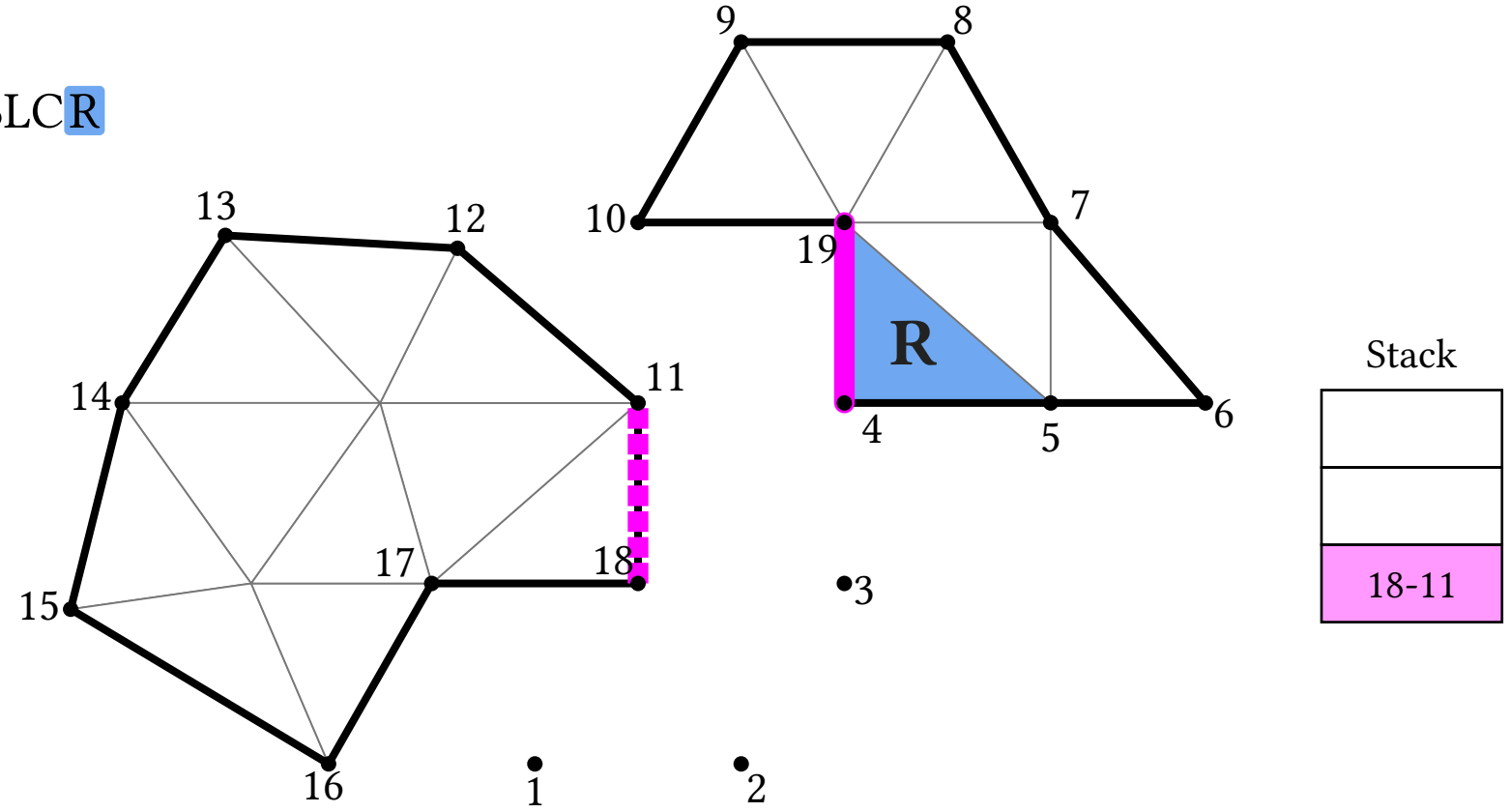


Figure 2: Jednoduchá komprese.

History = CCRRRSLCRS

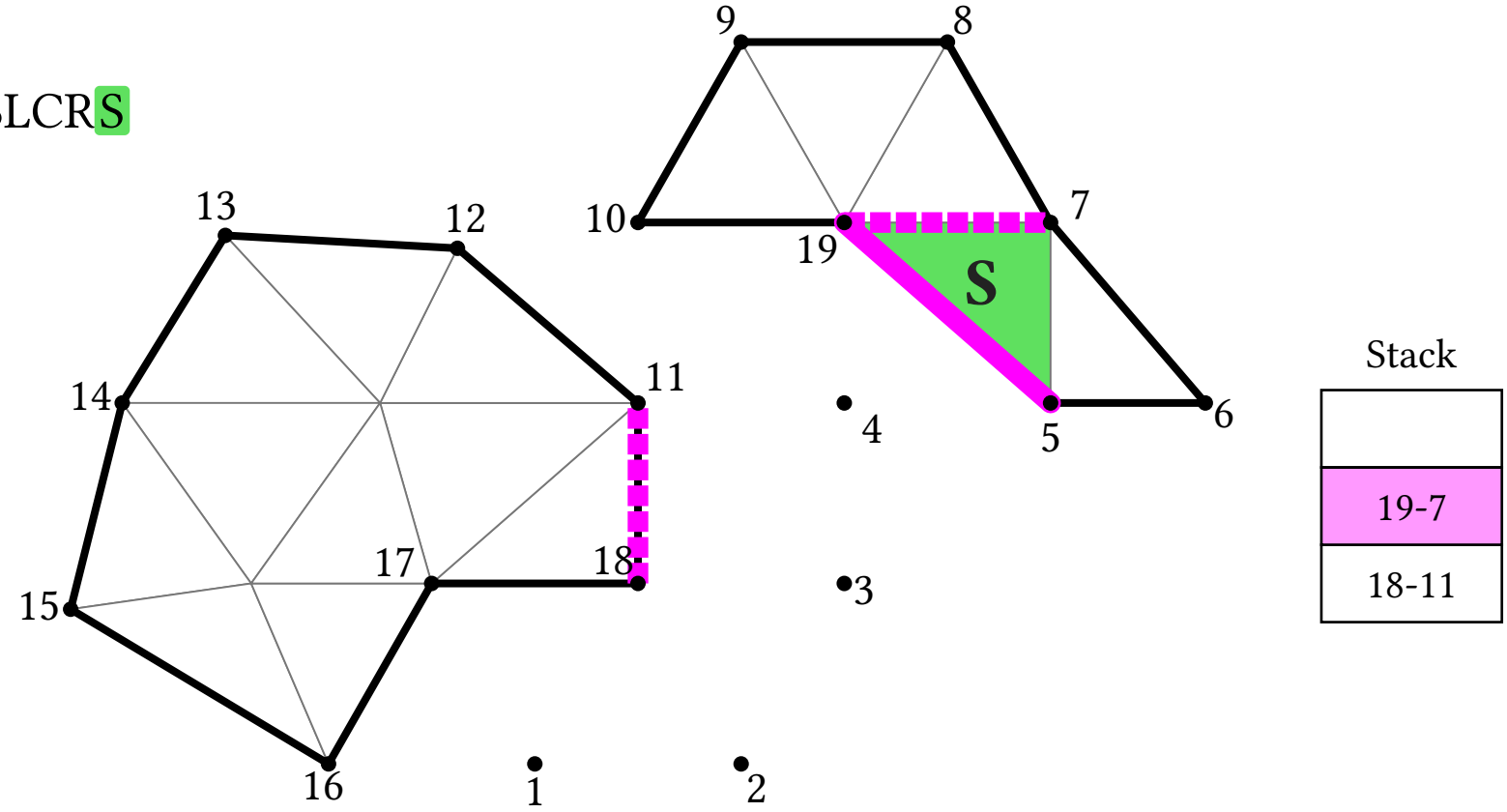


Figure 2: Jednoduchá komprese.

History = CCRRRSLCRSE

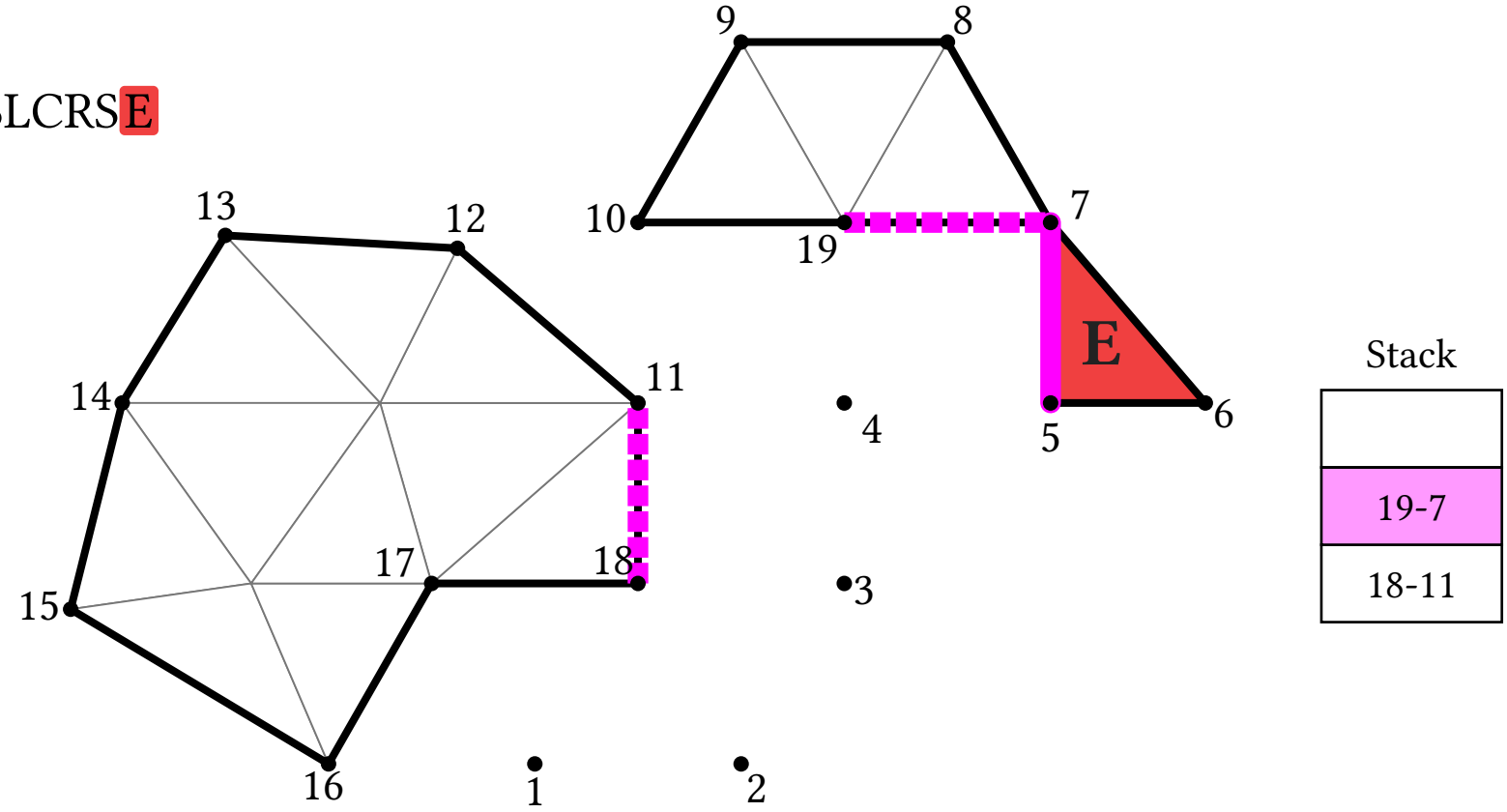


Figure 2: Jednoduchá komprese.

History = CCRRRSLCRSER

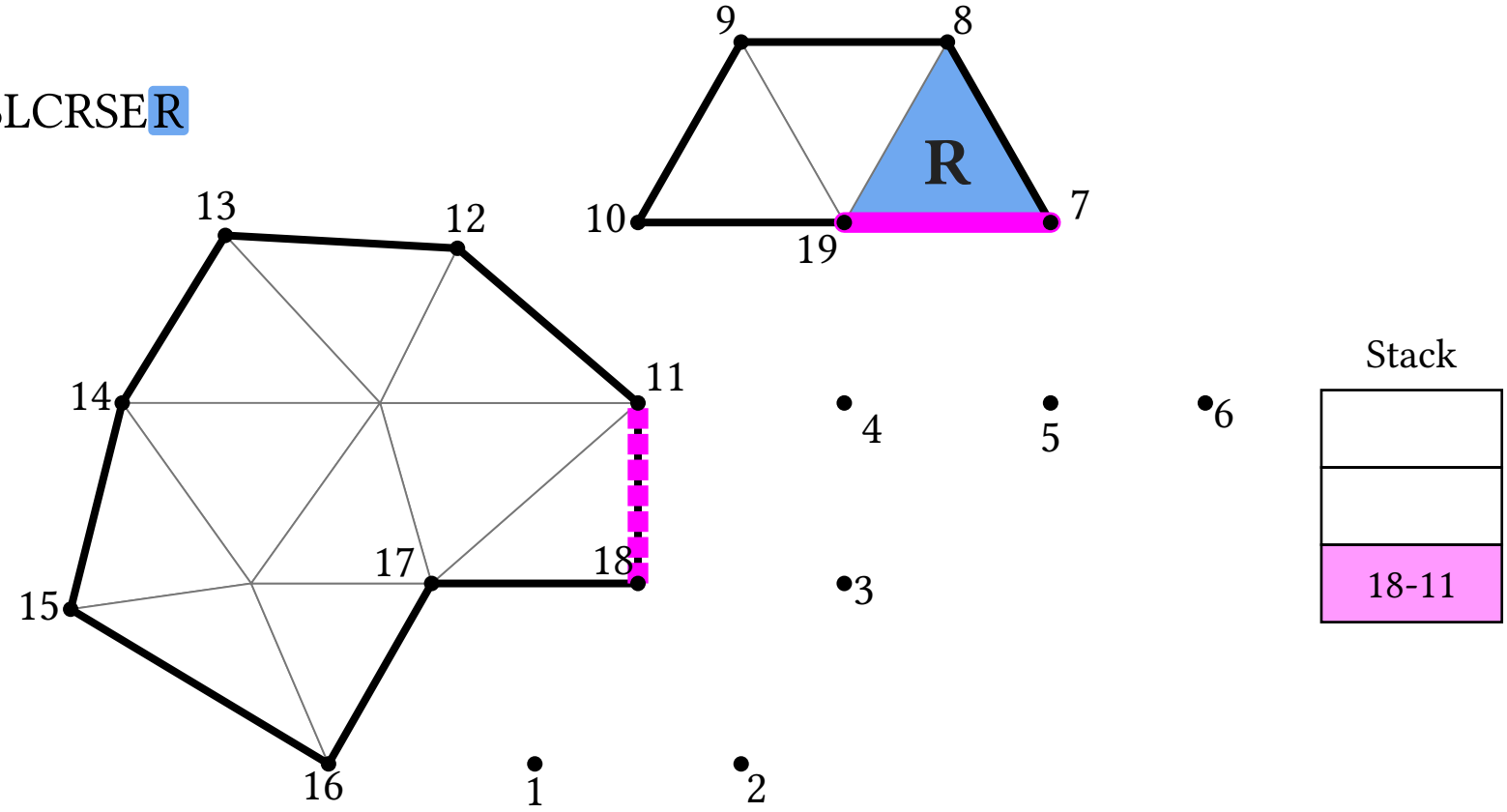


Figure 2: Jednoduchá komprese.

History = CCRRRSLCRSERR

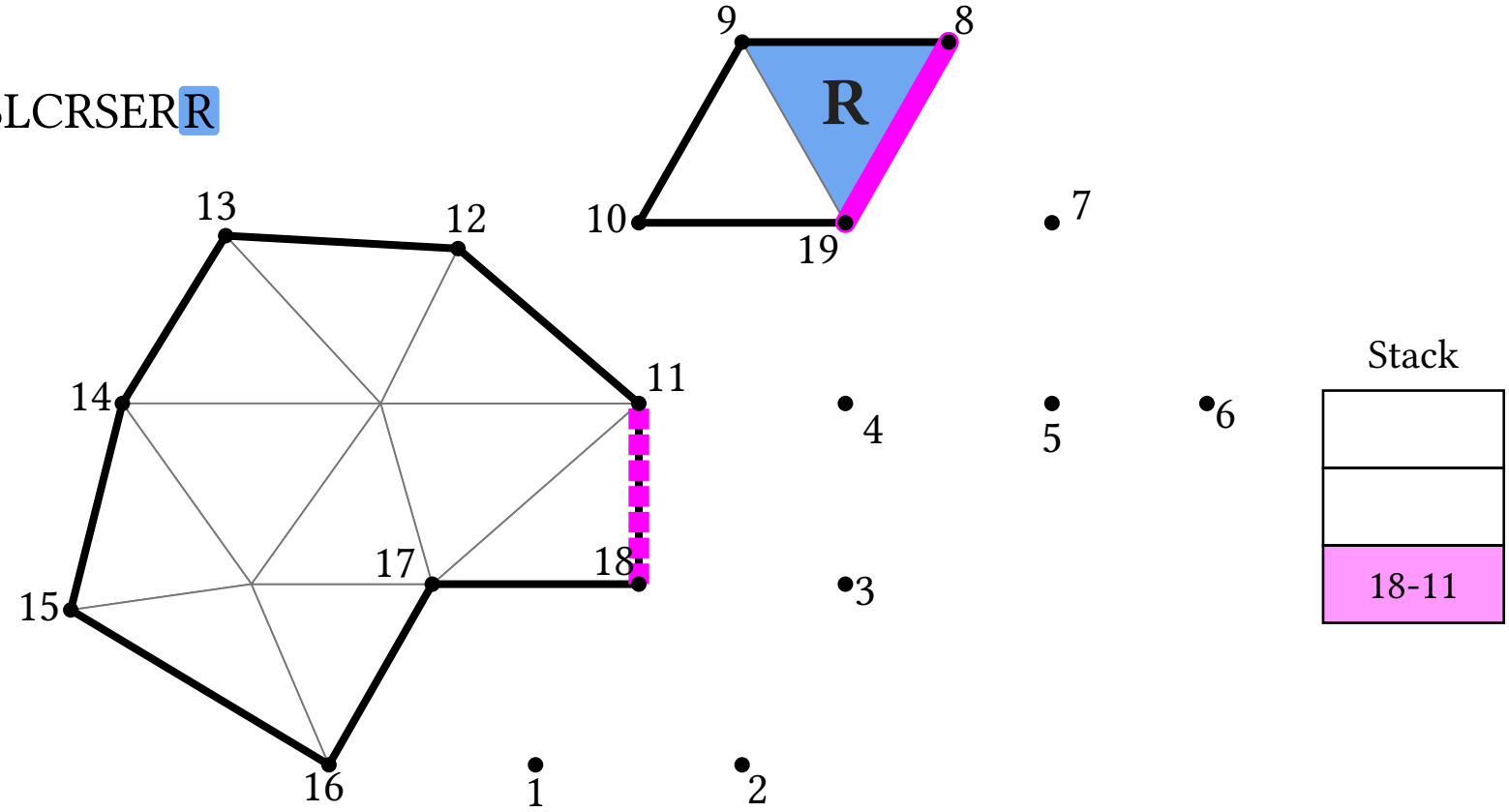


Figure 2: Jednoduchá komprese.

History = CCRRRSLCRSERRE

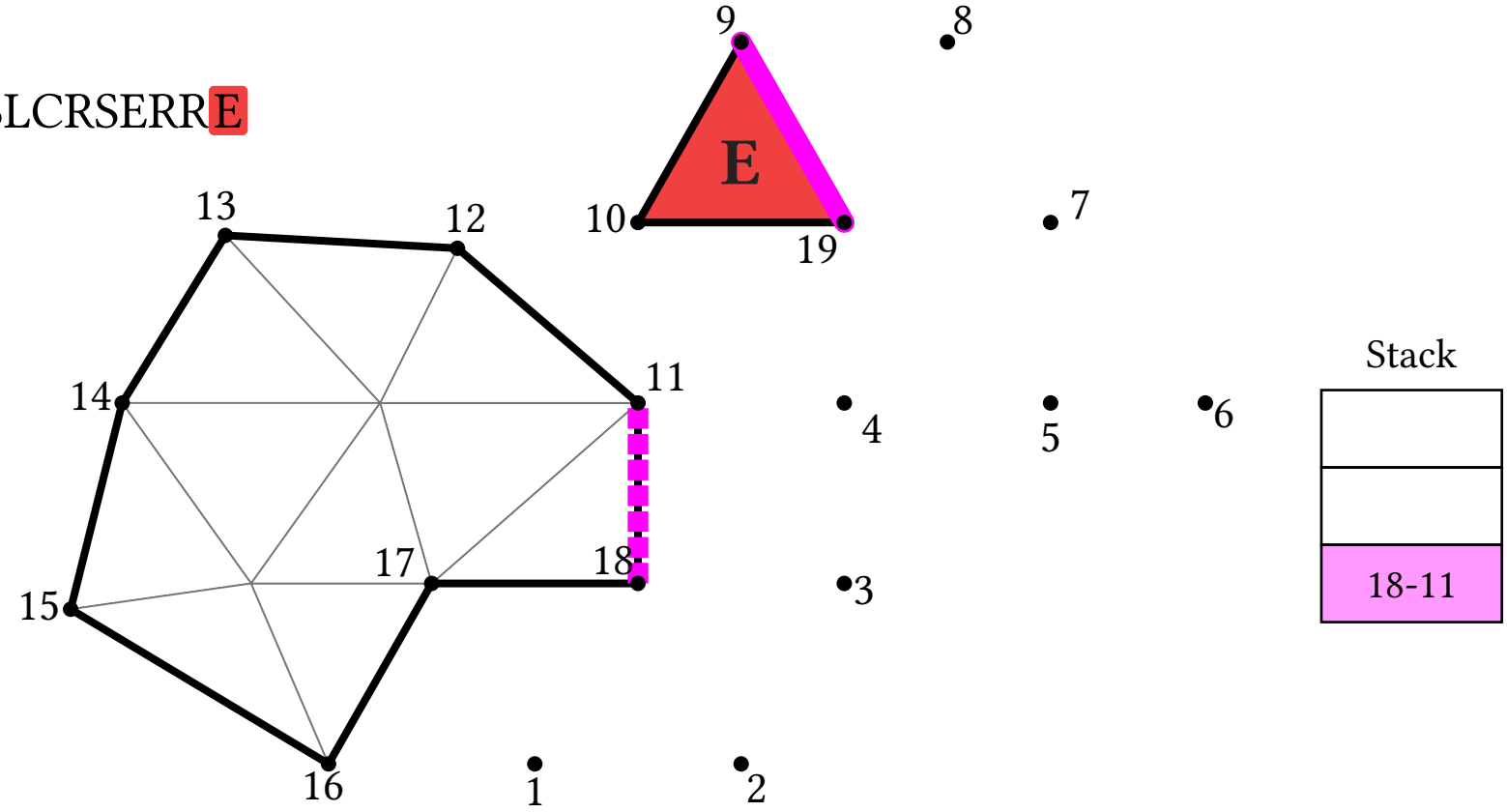


Figure 2: Jednoduchá komprese.

History = CCRRRSLCRSERREL

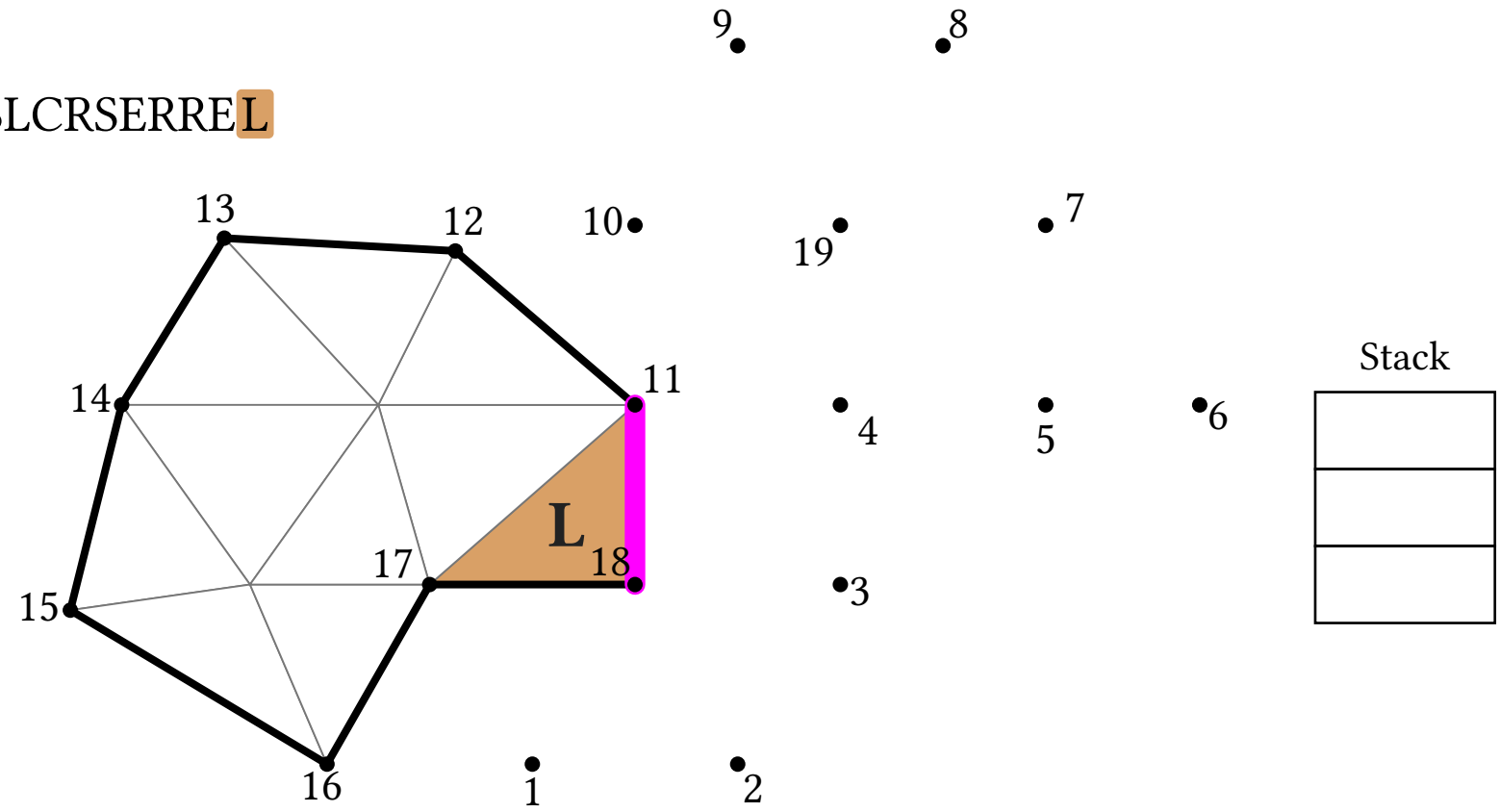


Figure 2: Jednoduchá komprese.

History = CCRRRSLCRSERREL**C**

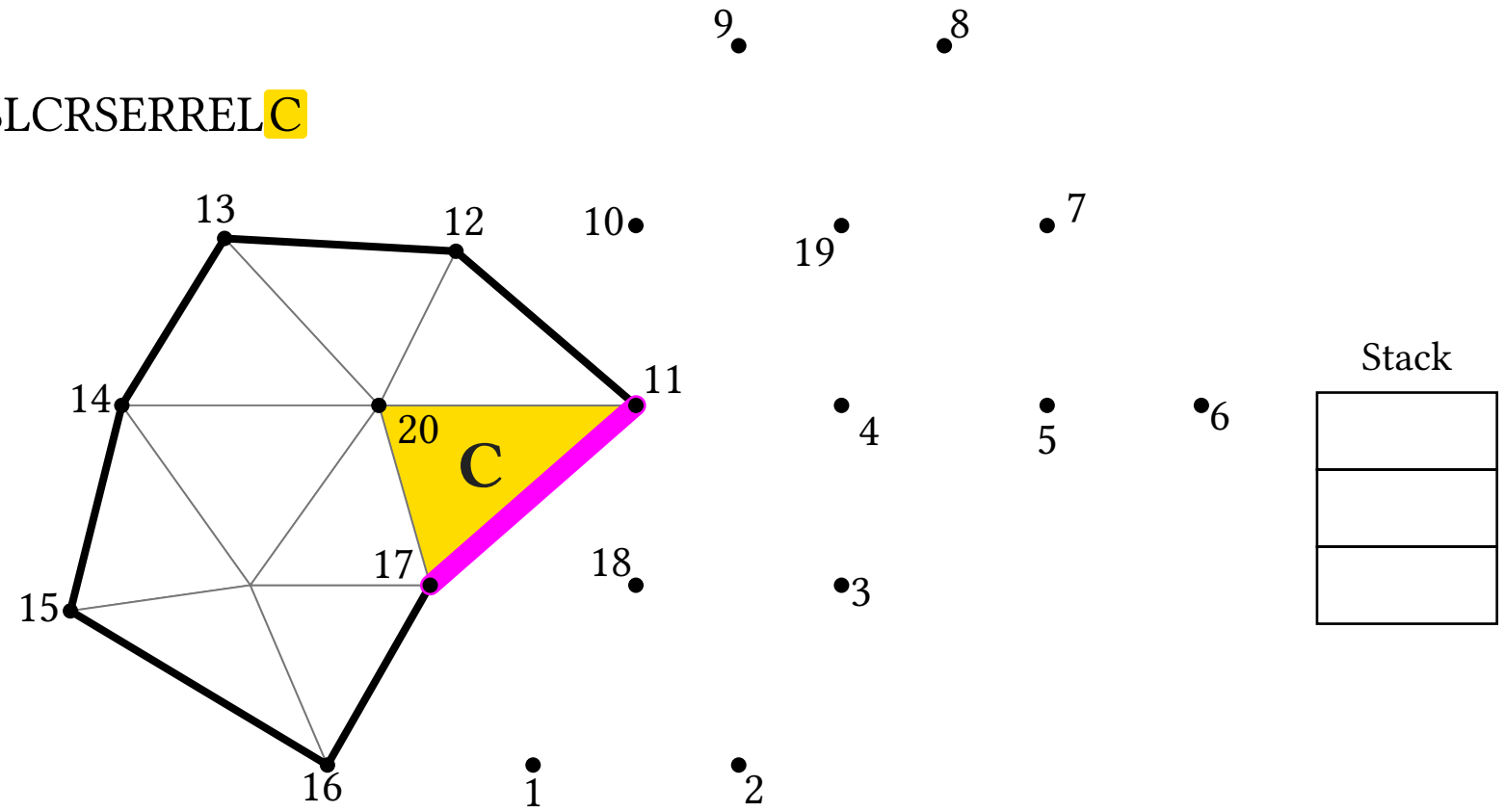


Figure 2: Jednoduchá komprese.

History = CCRRRSLCRSERRELCR

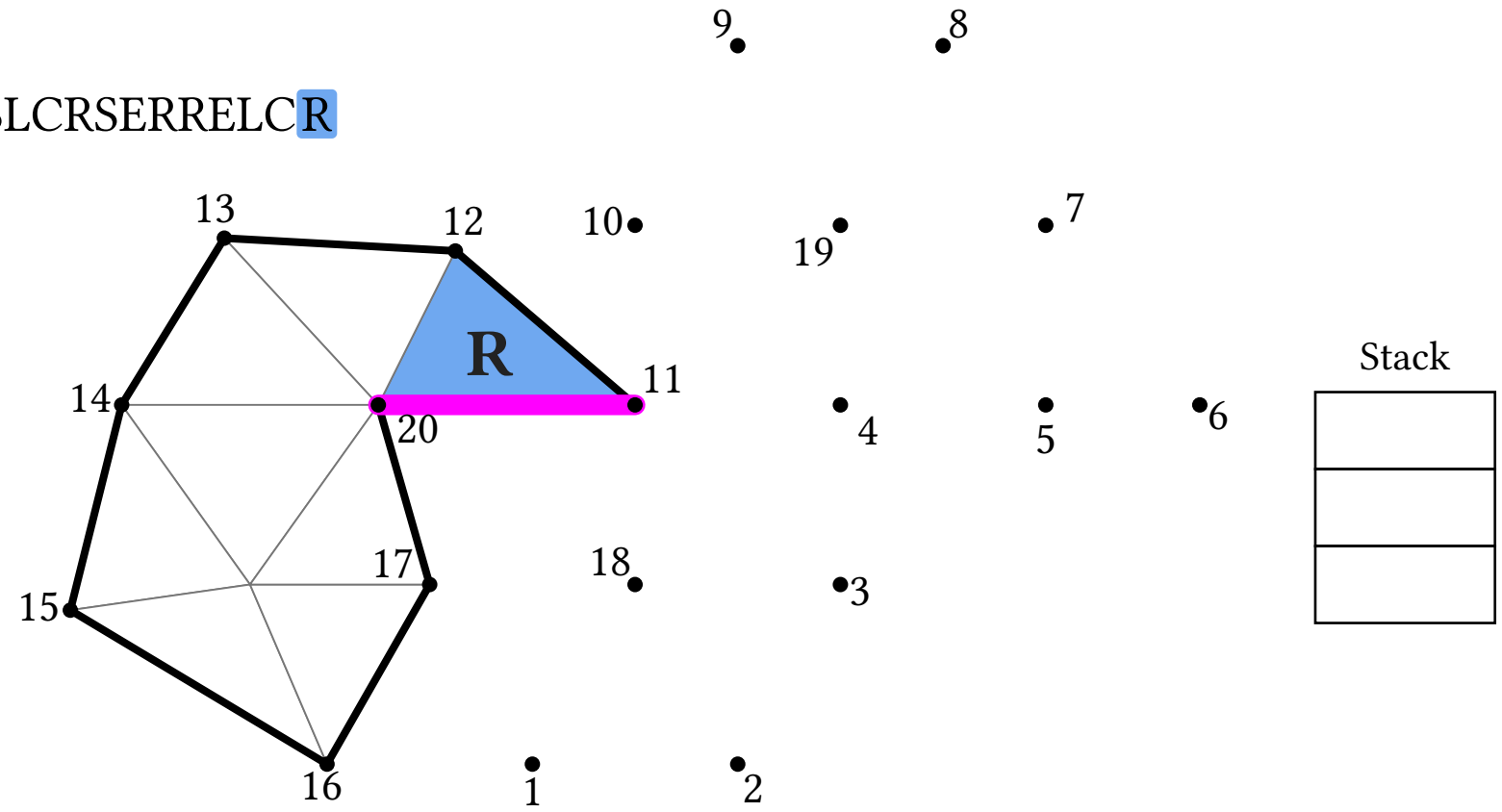


Figure 2: Jednoduchá komprese.

History = CCRRRSLCRSERRELCRR

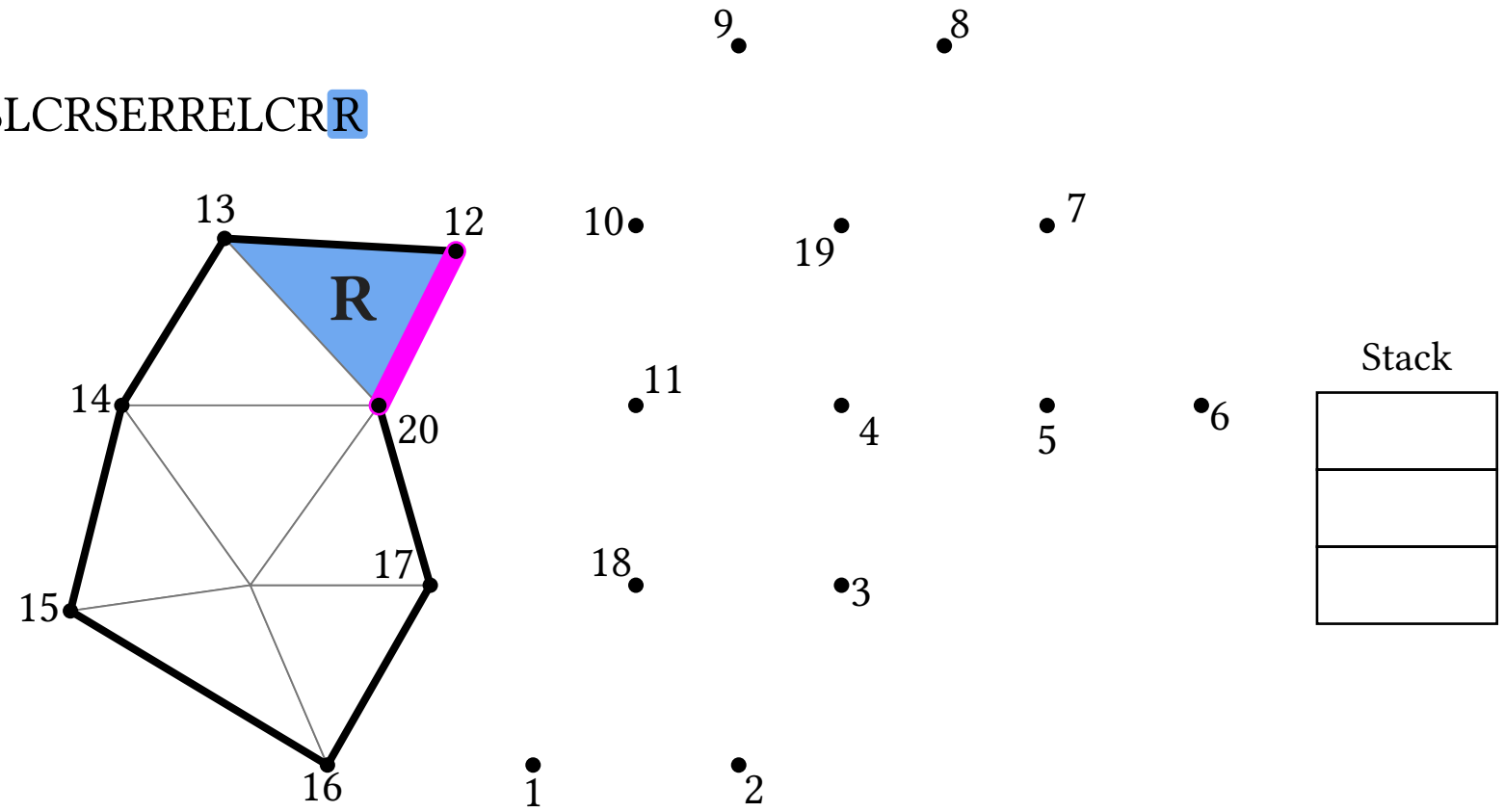


Figure 2: Jednoduchá komprese.

History = CCRRRSLCRSERRELCRRR

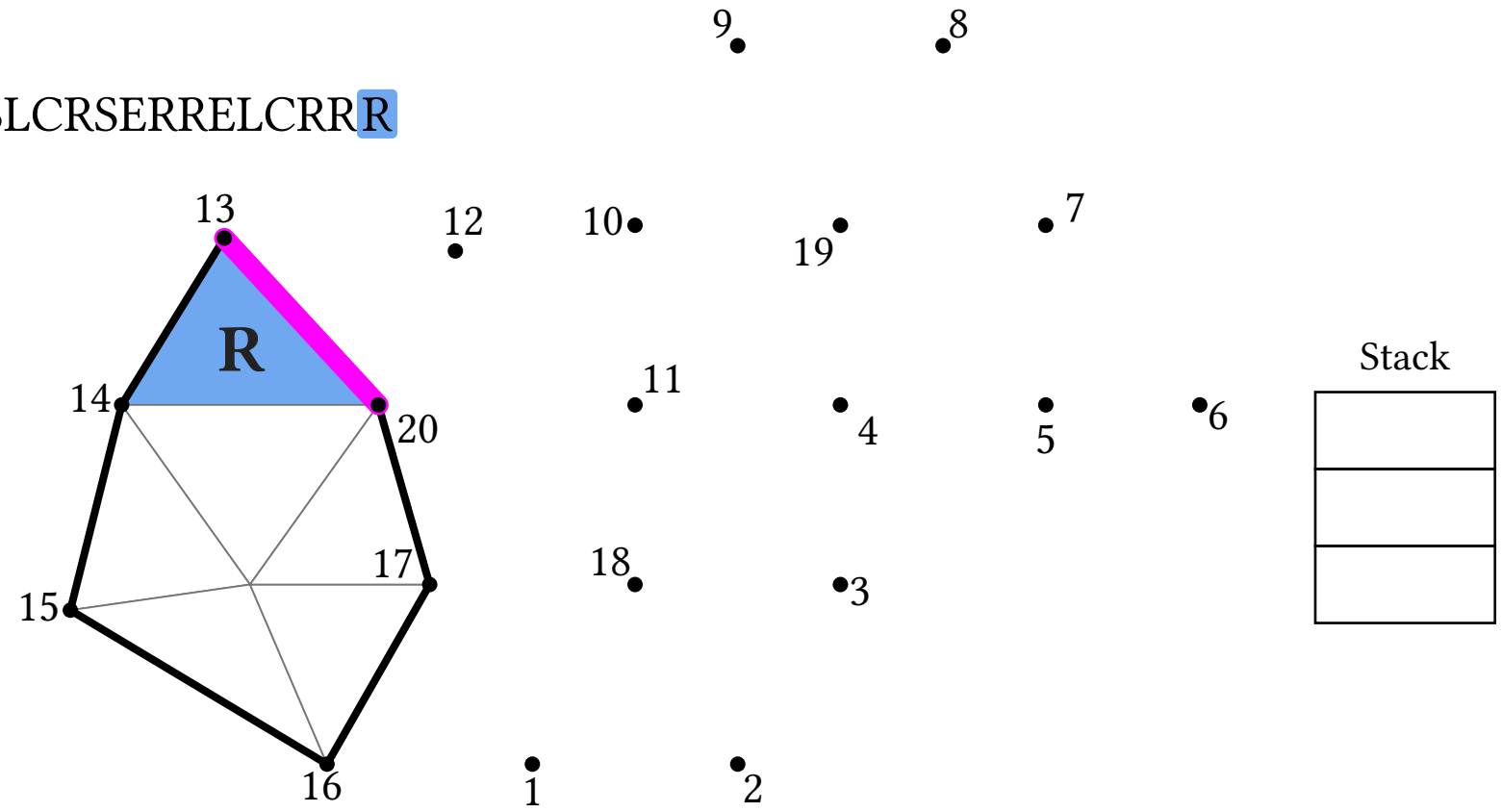


Figure 2: Jednoduchá komprese.

History = CCRRRSLCRSERRELCRRRC

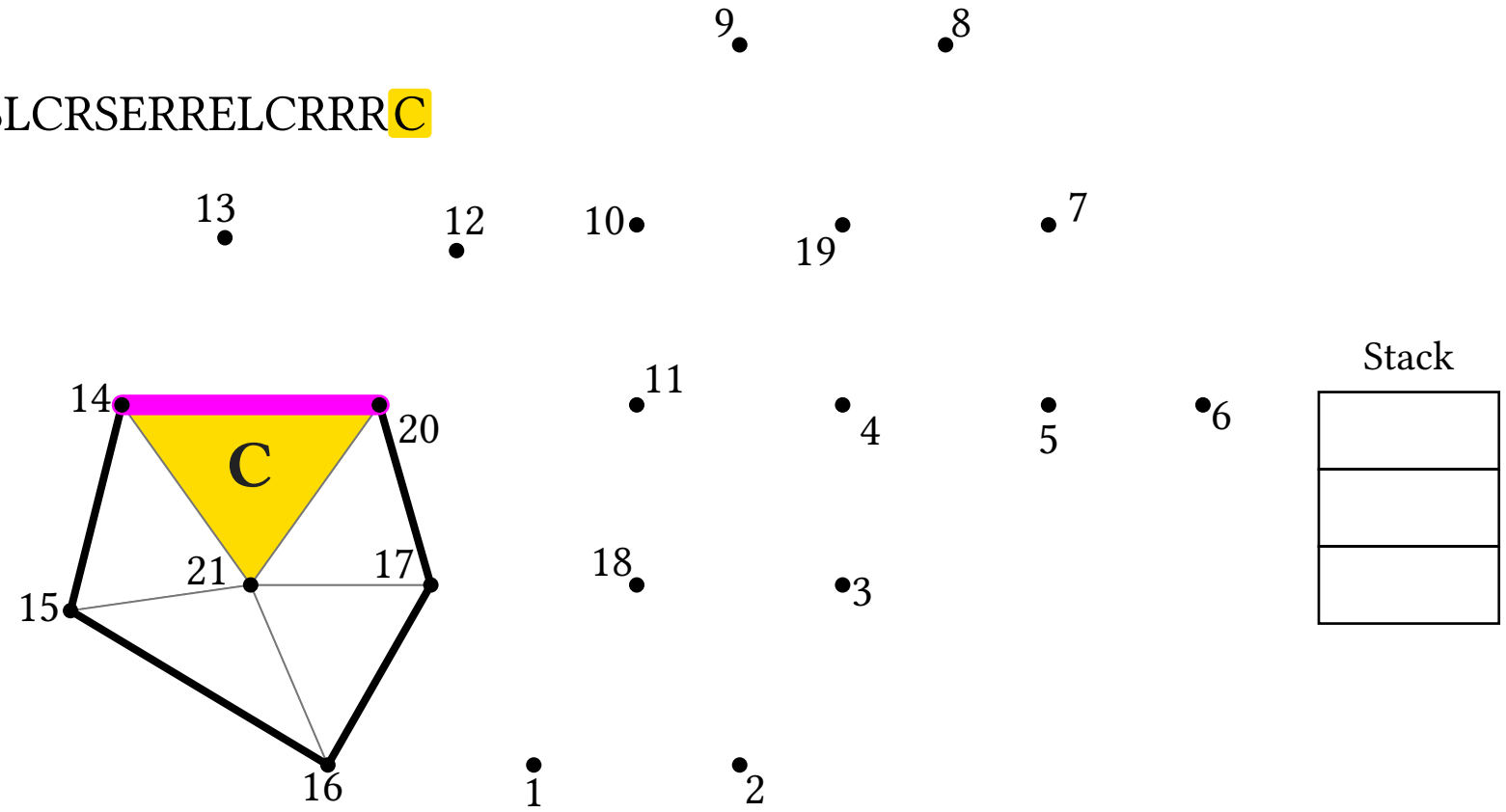


Figure 2: Jednoduchá komprese.

History = CCRRRSLCRSERRELCRRRCR

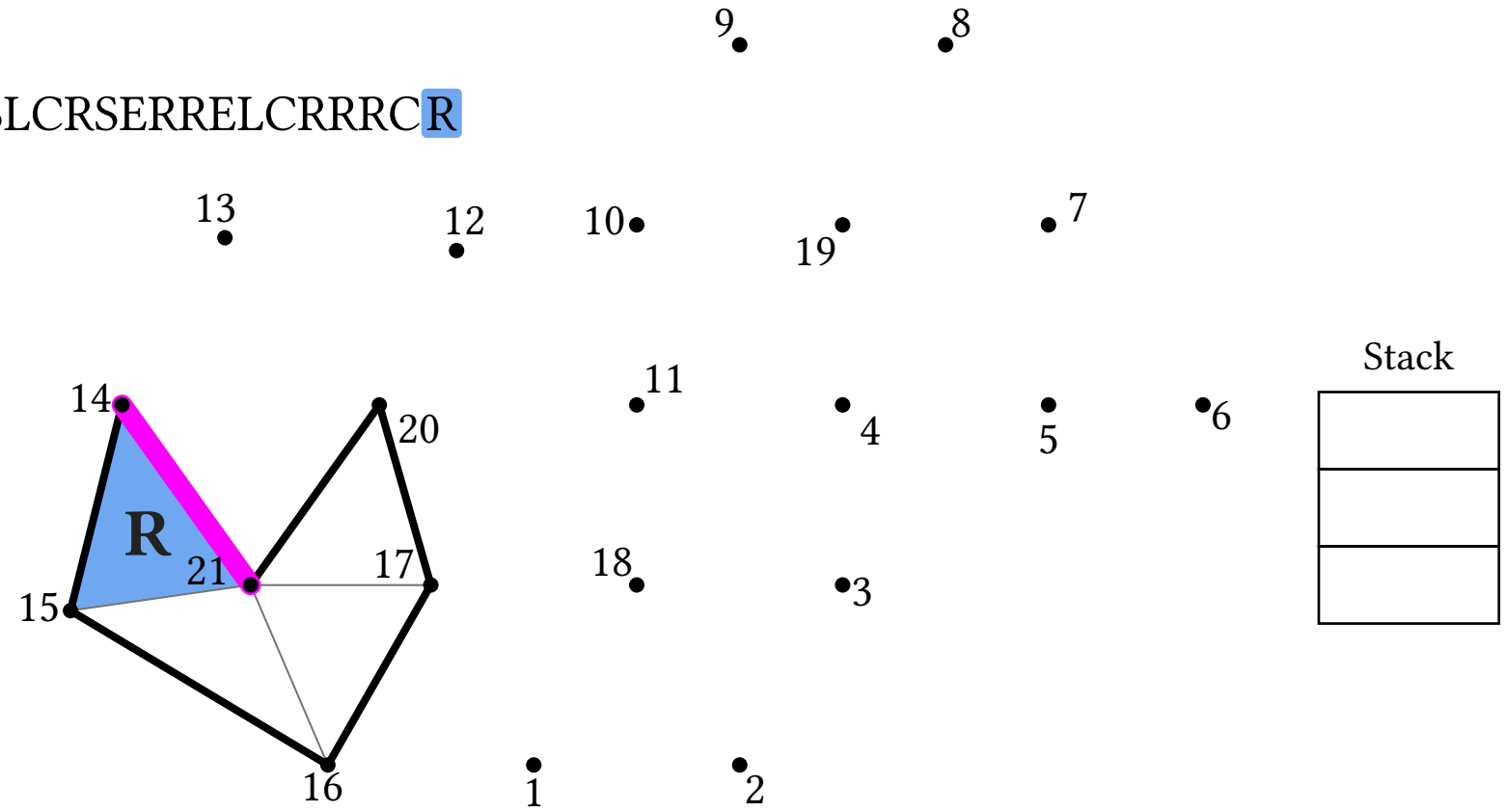


Figure 2: Jednoduchá komprese.

History = CCRRRSLCRSERRELCRRRCRR

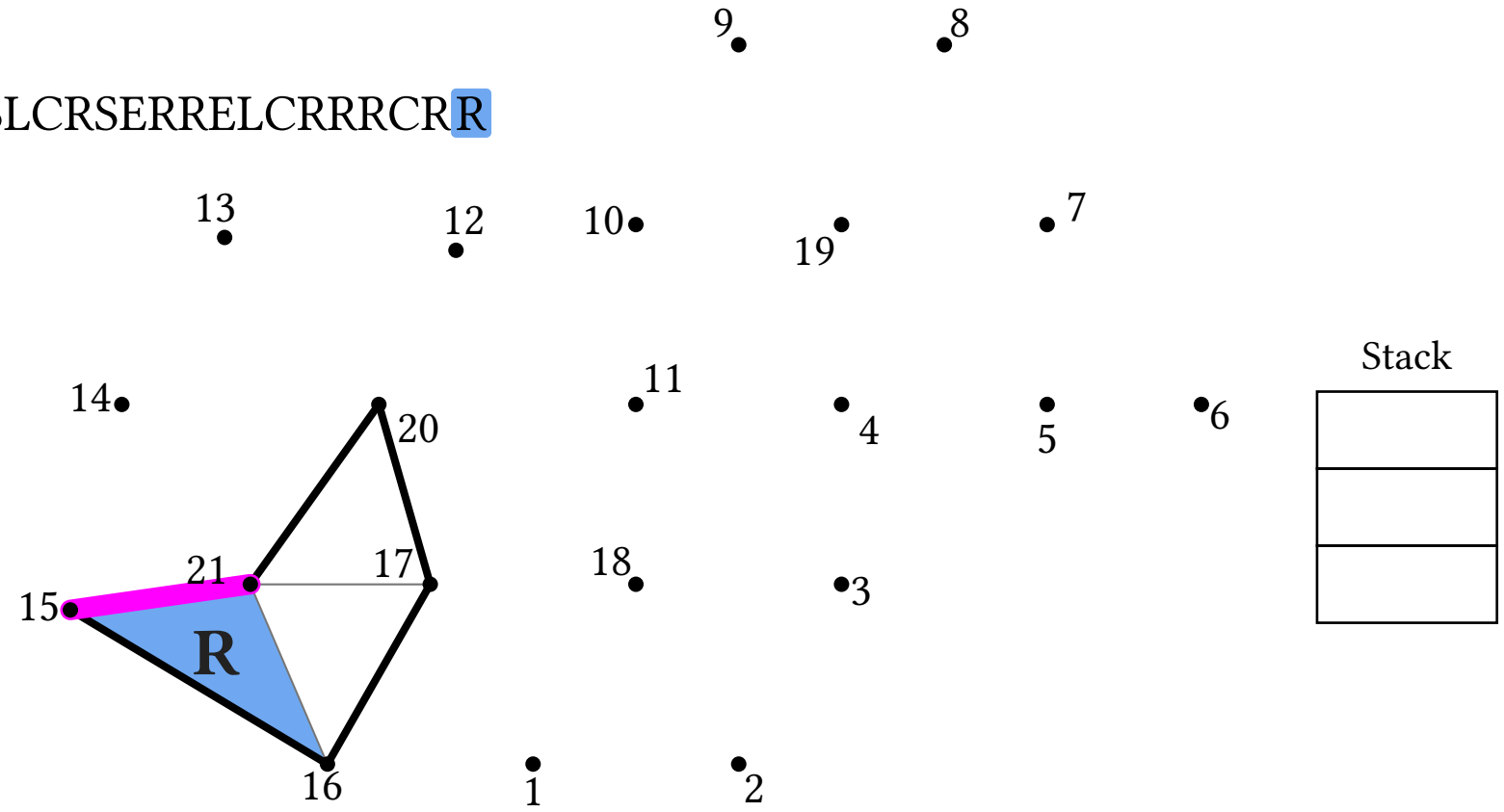


Figure 2: Jednoduchá komprese.

History = CCRRRSLCRSERRELCRRRCRRR

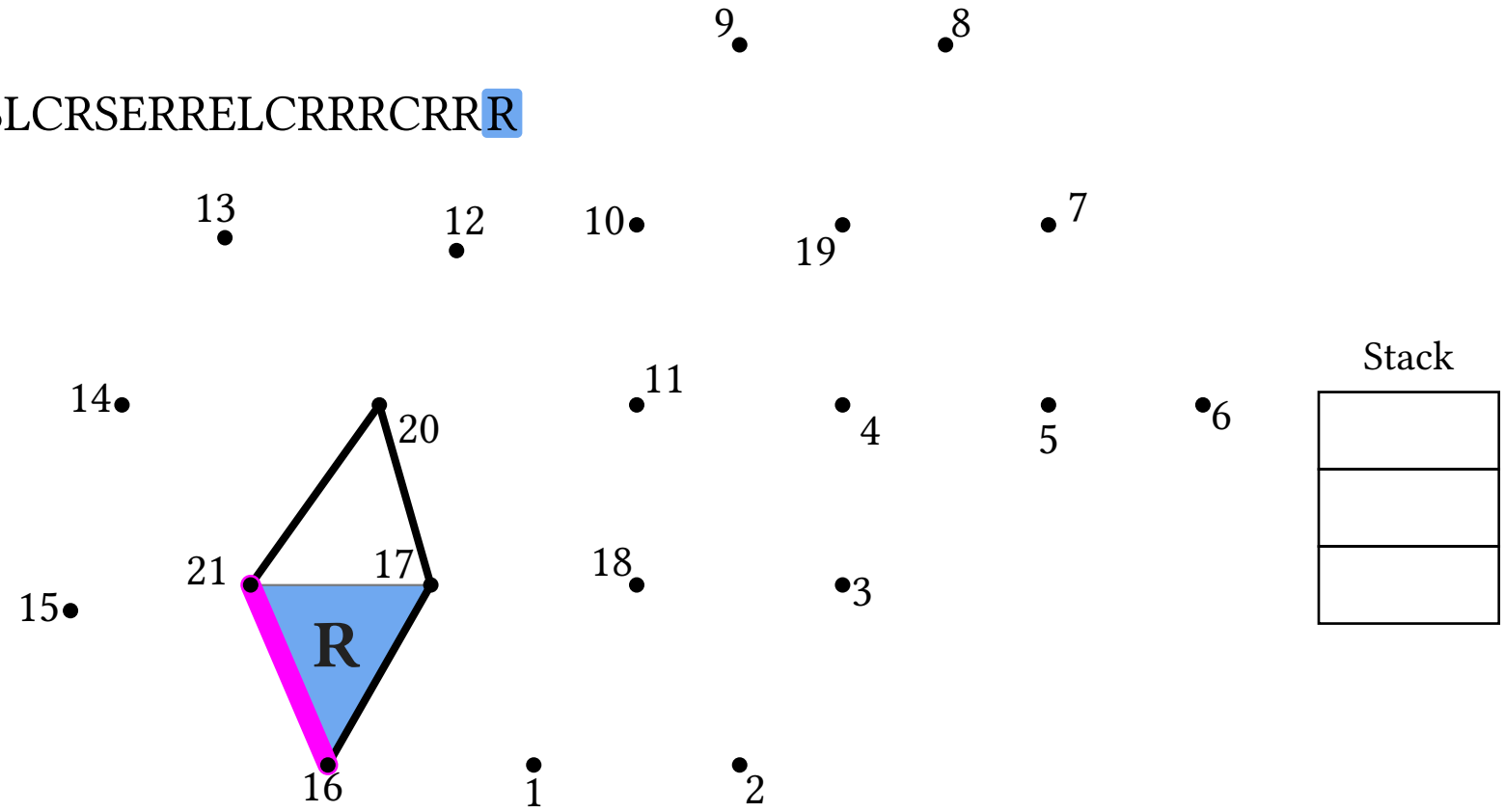


Figure 2: Jednoduchá komprese.

History = CCRRRSLCRSERRELCRRRCRRRE

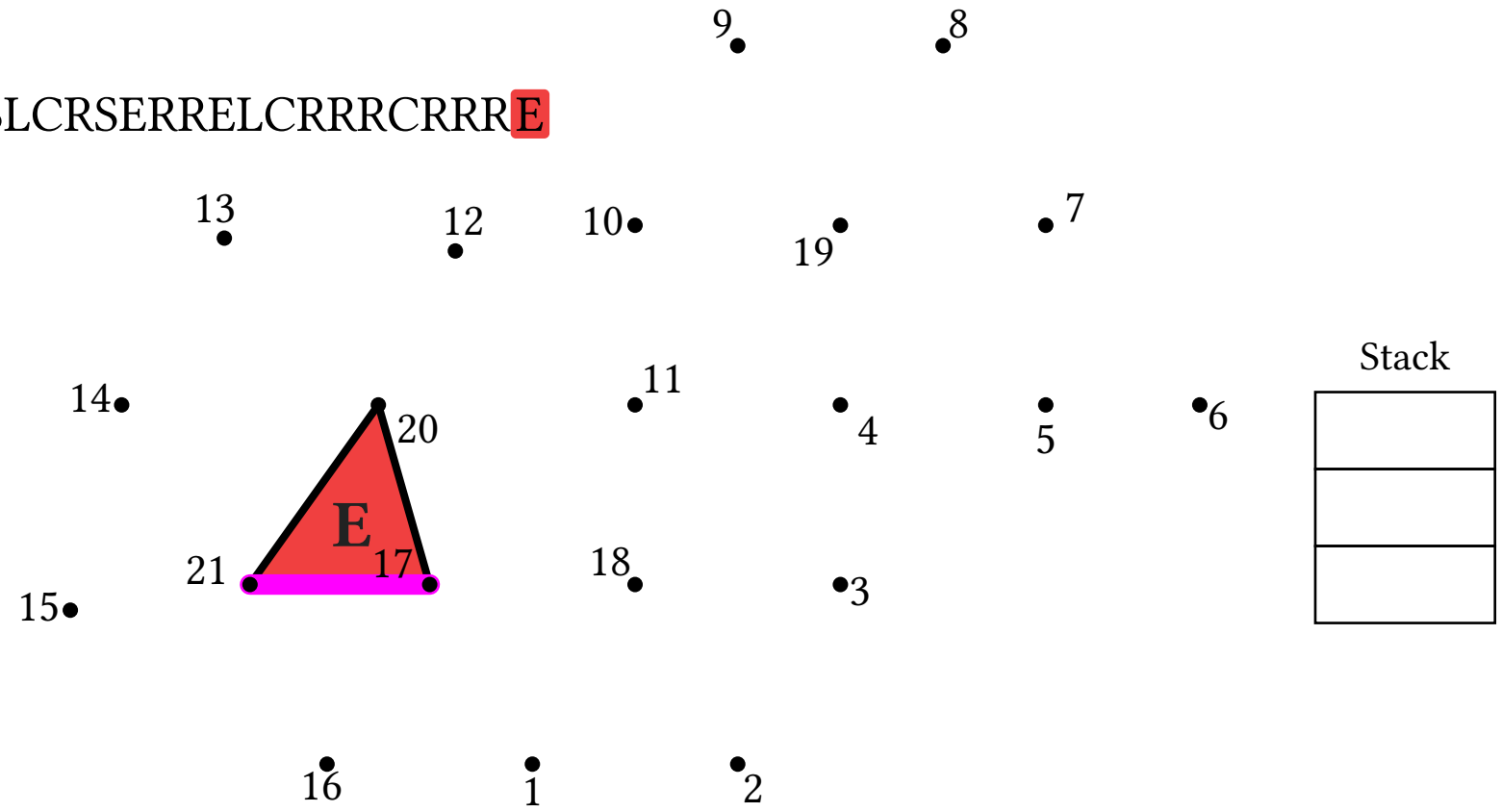


Figure 2: Jednoduchá komprese.

History = CCRRRSLCRSERRELCRRRCRRRE

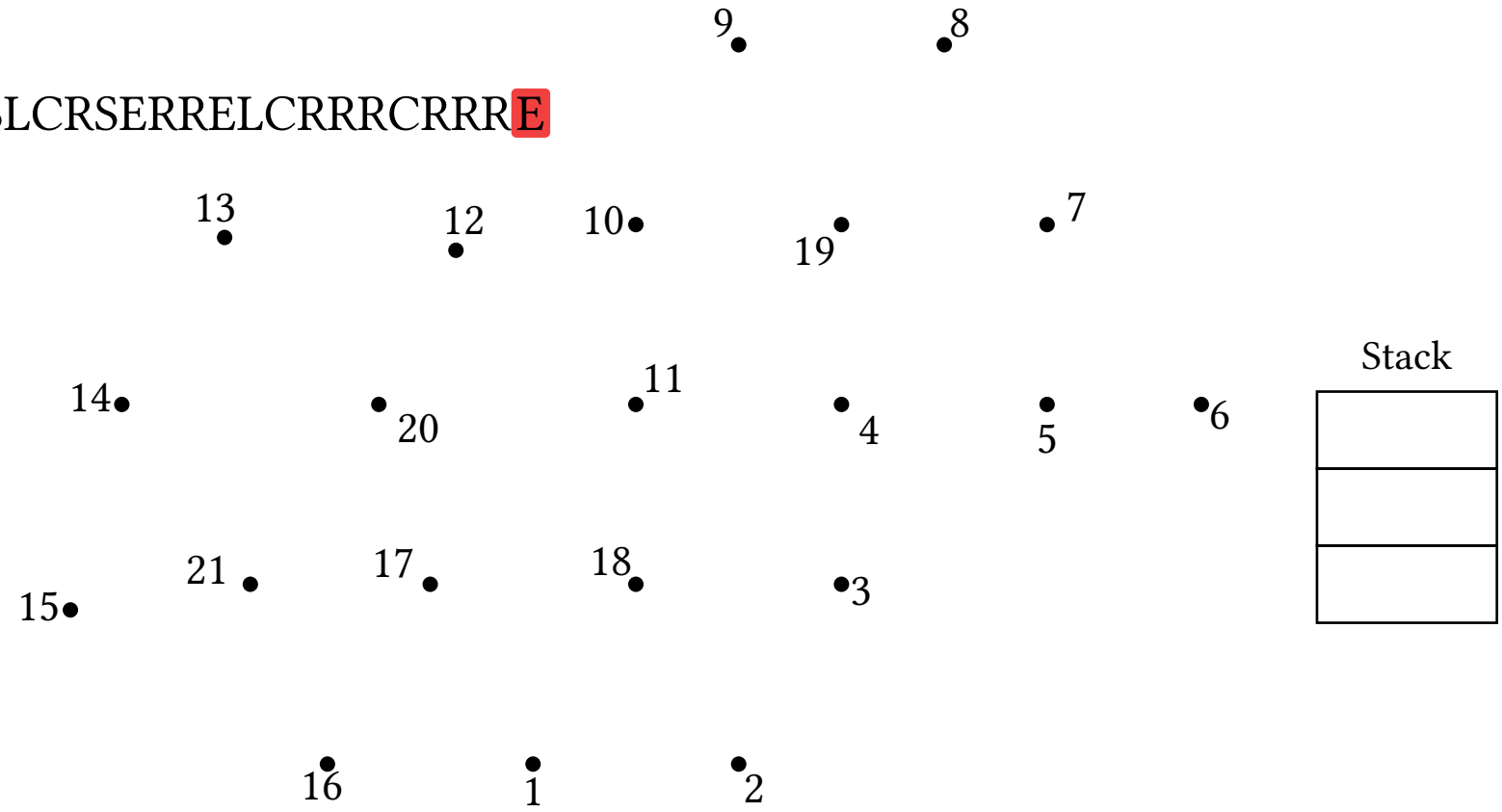


Figure 2: Jednoduchá komprese.

2.3 Vyšší genus?

History =

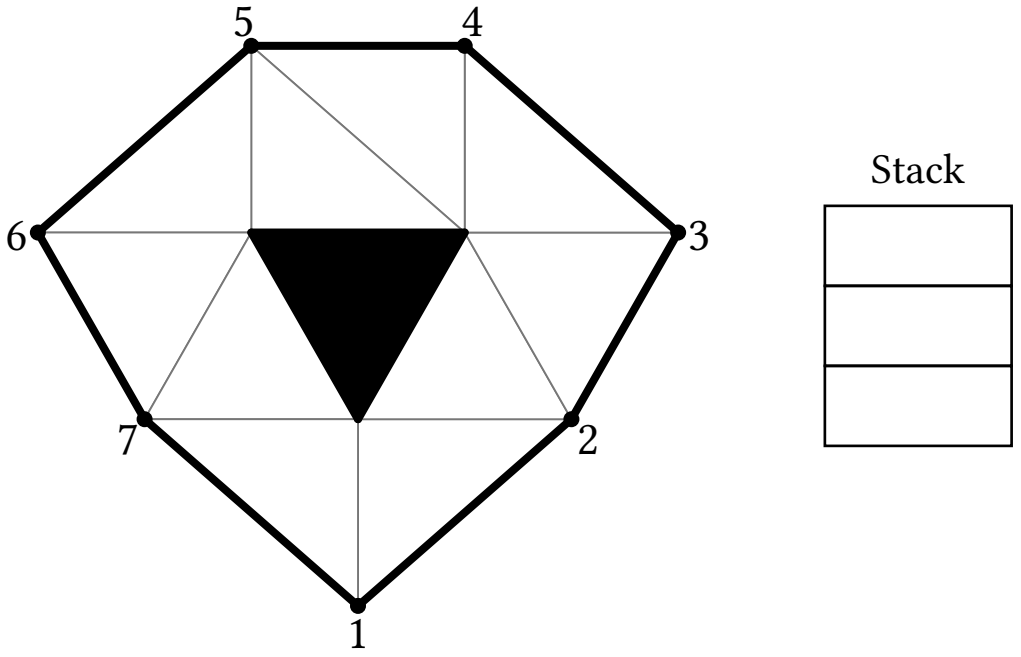


Figure 3: Komprese díry.

2.3 Vyšší genus?

History = S

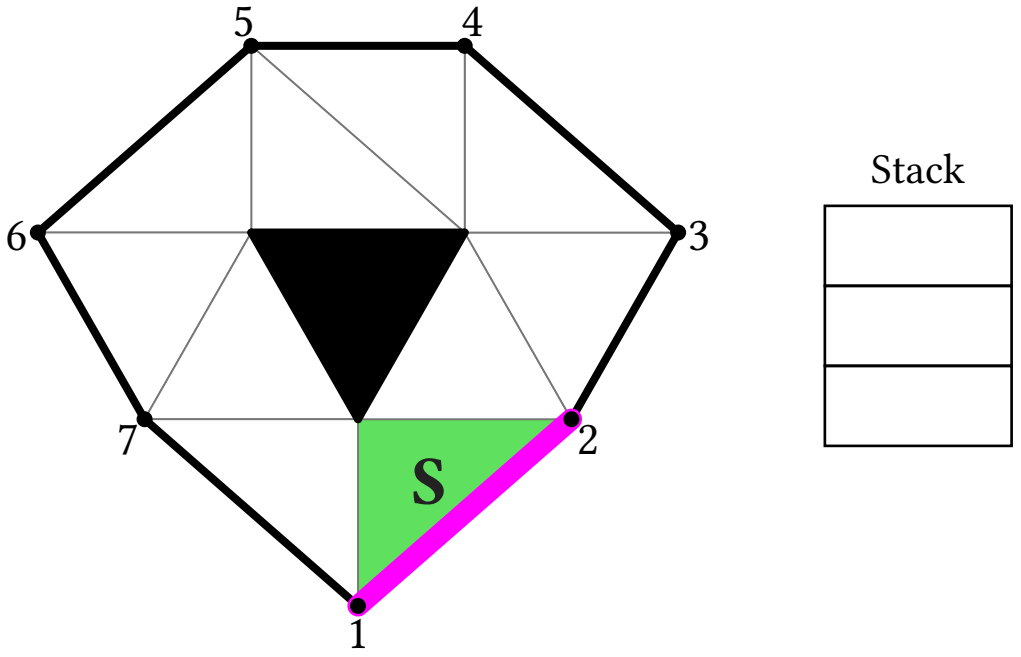
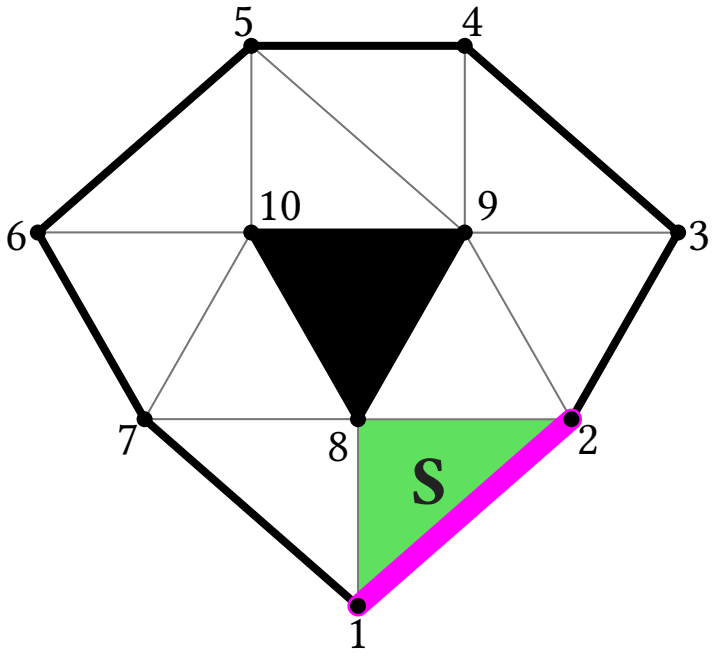


Figure 3: Komprese díry.

2.3 Vyšší genus?

History = S



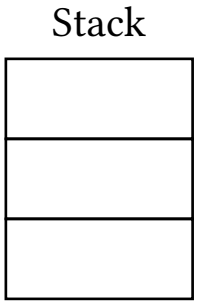


Figure 3: Komprese díry.

2.3 Vyšší genus?

History = SL

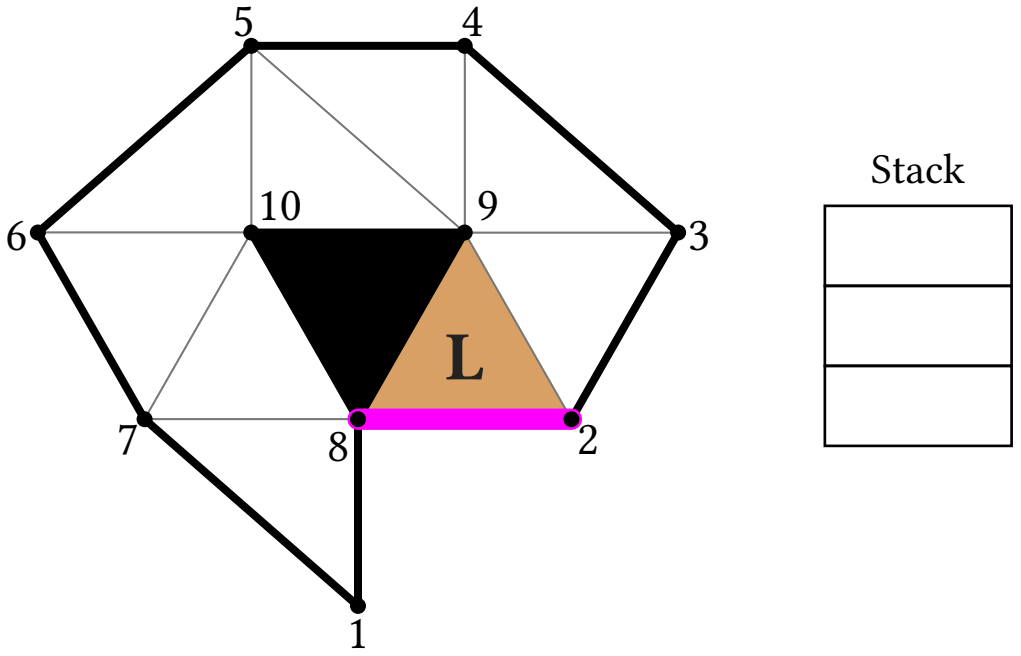


Figure 3: Komprese díry.

2.3 Vyšší genus?

History = SLR

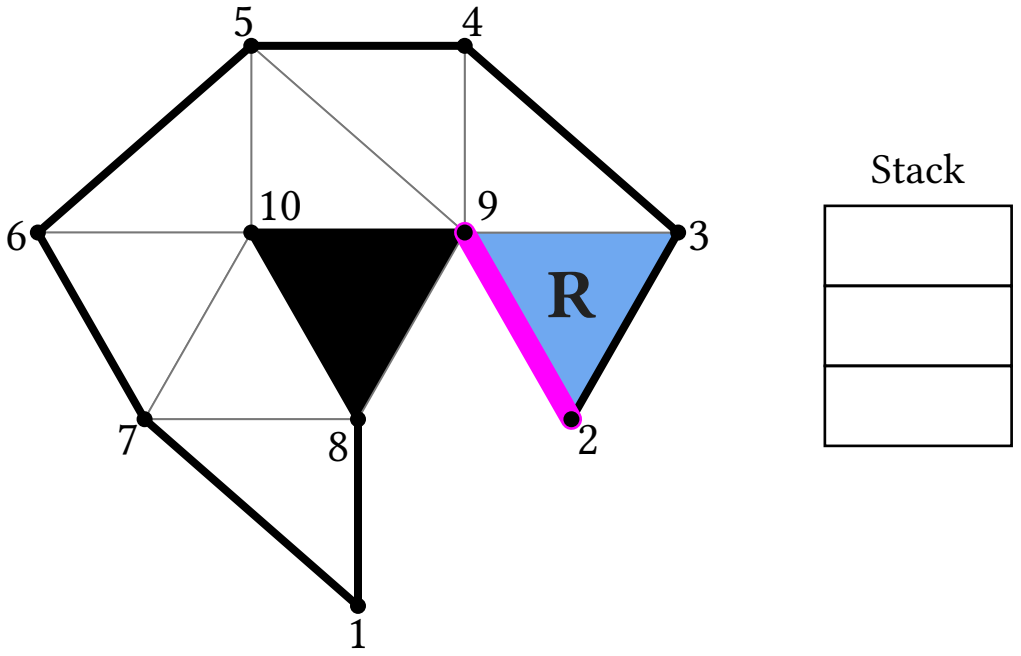


Figure 3: Komprese díry.

2.3 Vyšší genus?

History = SLRR

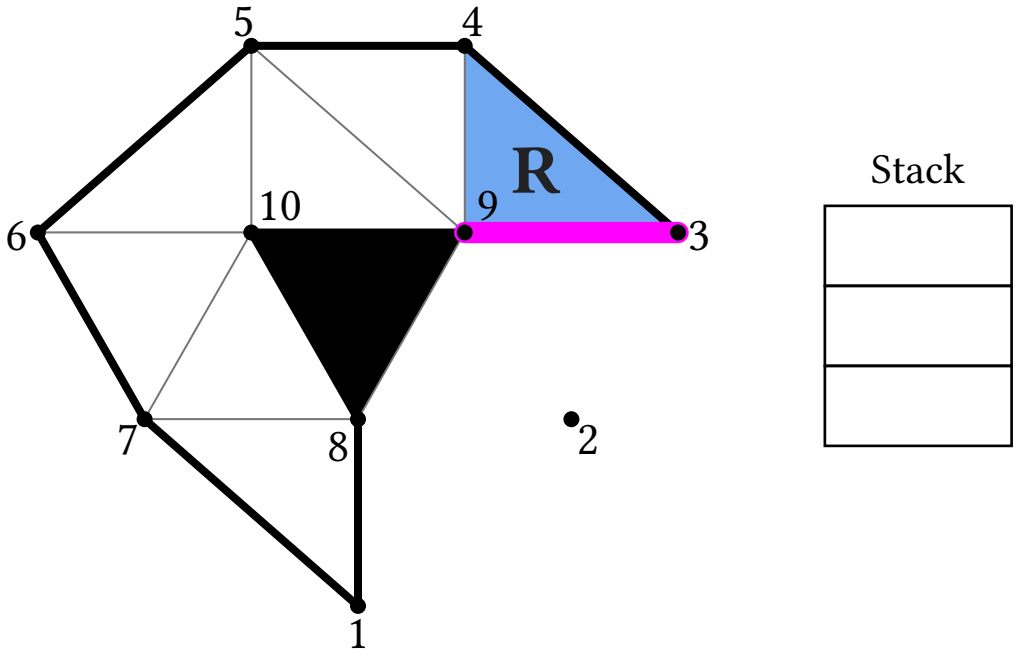


Figure 3: Komprese díry.

2.3 Vyšší genus?

History = SLRRR

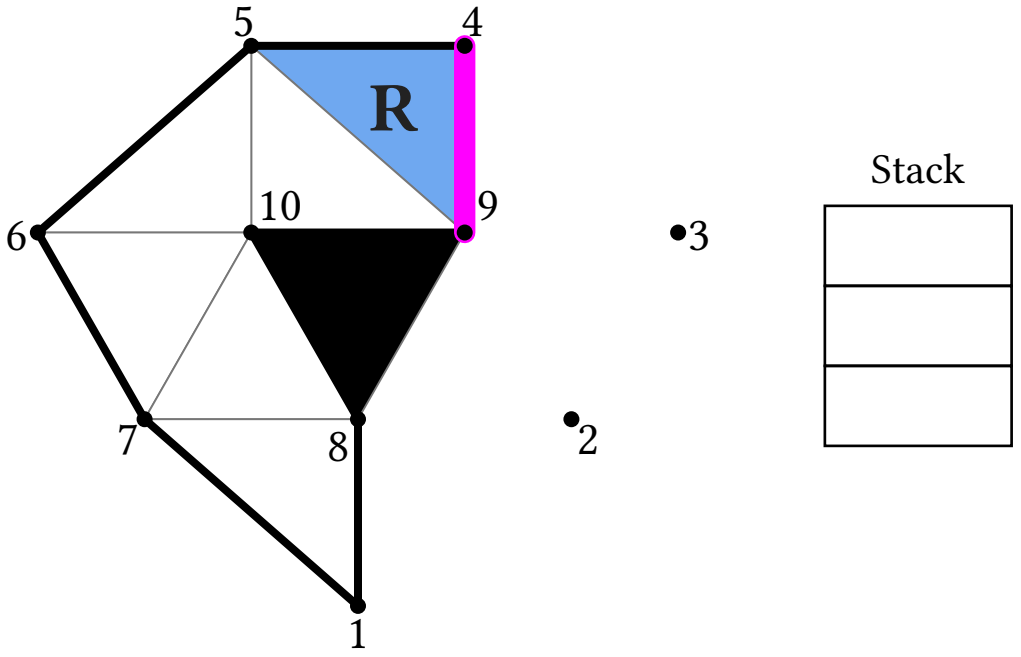


Figure 3: Komprese díry.

2.3 Vyšší genus?

History = SLRRRL

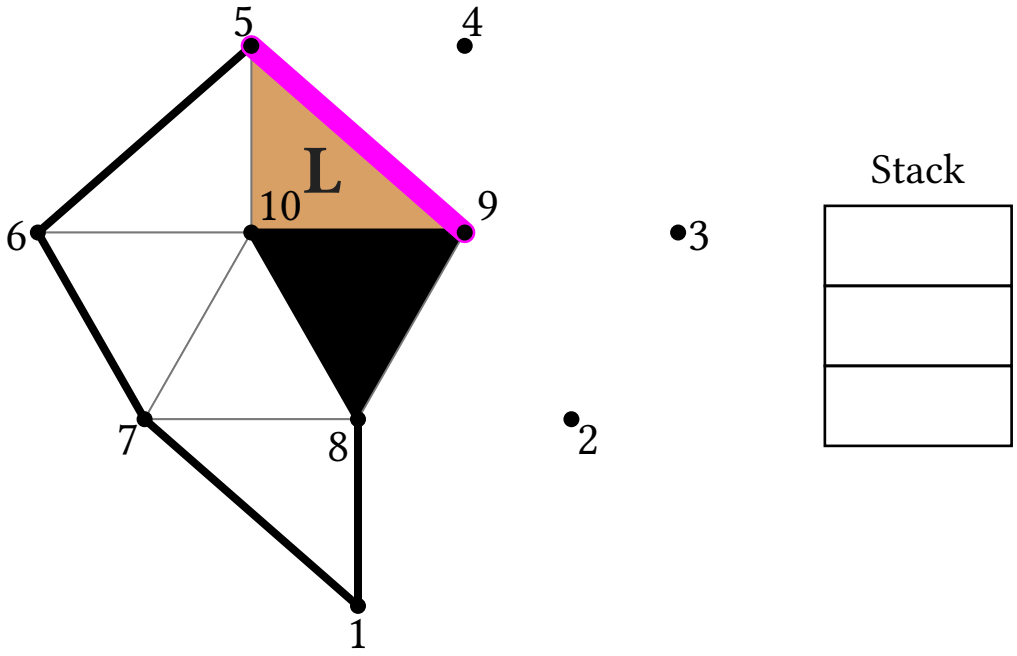


Figure 3: Komprese díry.

2.3 Vyšší genus?

History = SLRRRLR

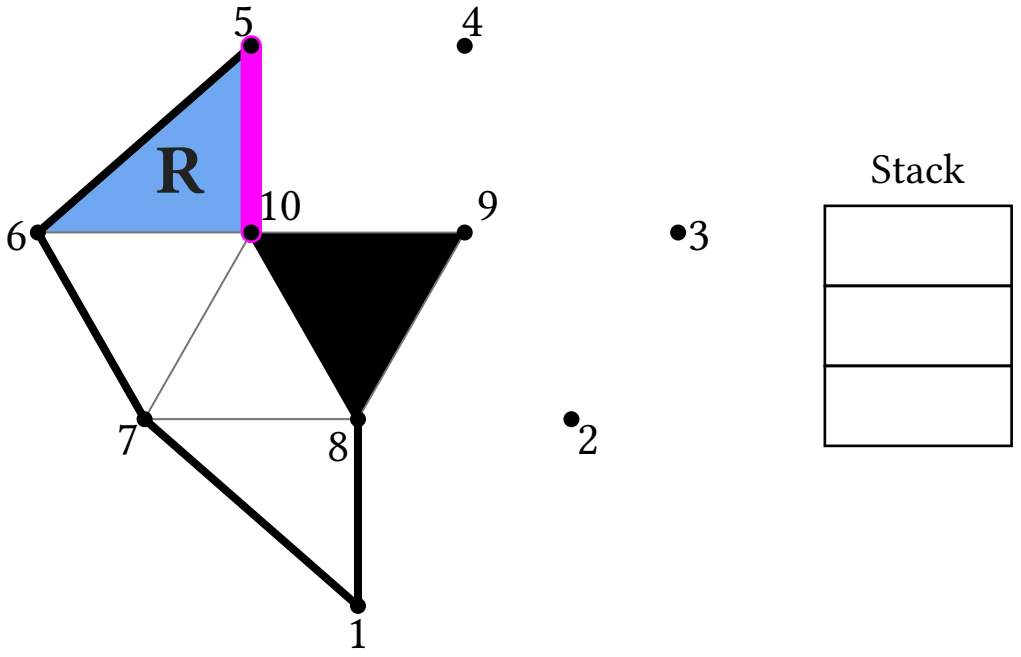


Figure 3: Komprese díry.

2.3 Vyšší genus?

History = SLRRRLRR

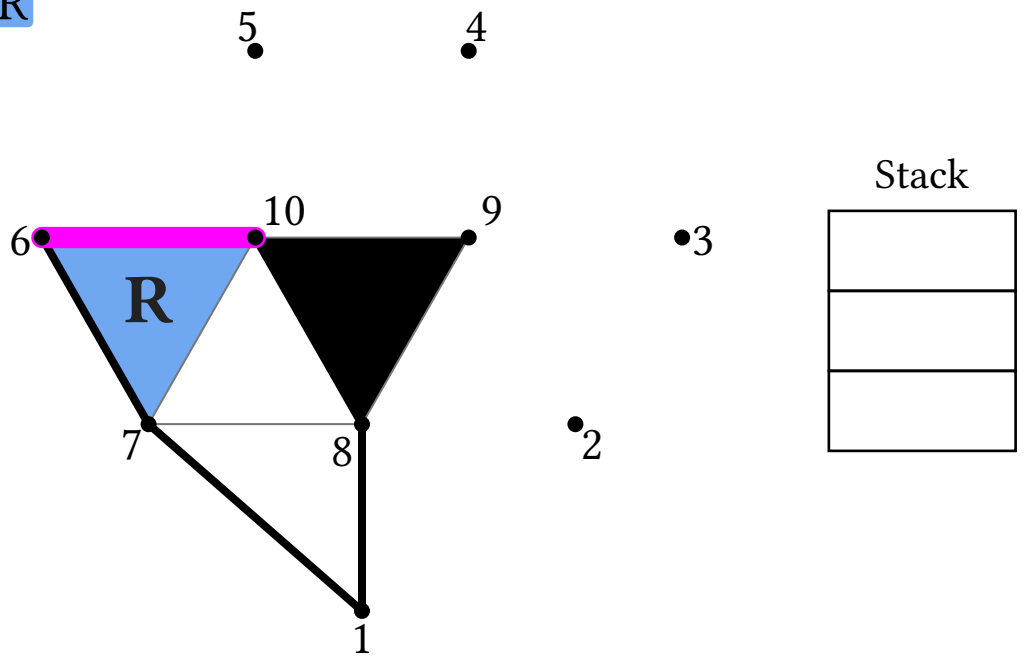


Figure 3: Komprese díry.

2.3 Vyšší genus?

History = SLRRRLRRL

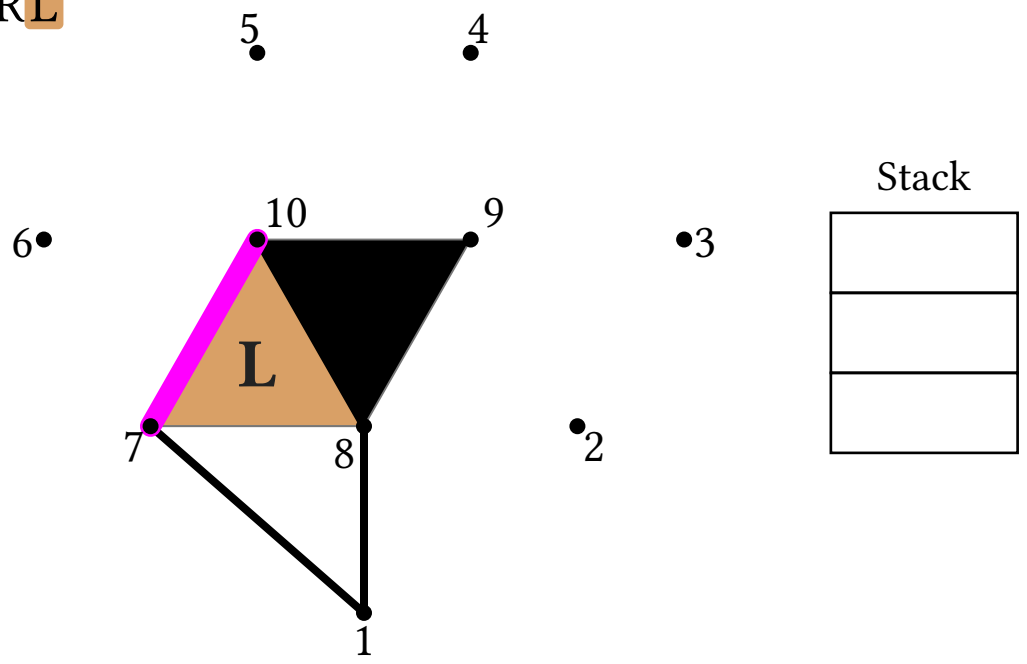


Figure 3: Komprese díry.

2.3 Vyšší genus?

History = SLRRRLRRLE

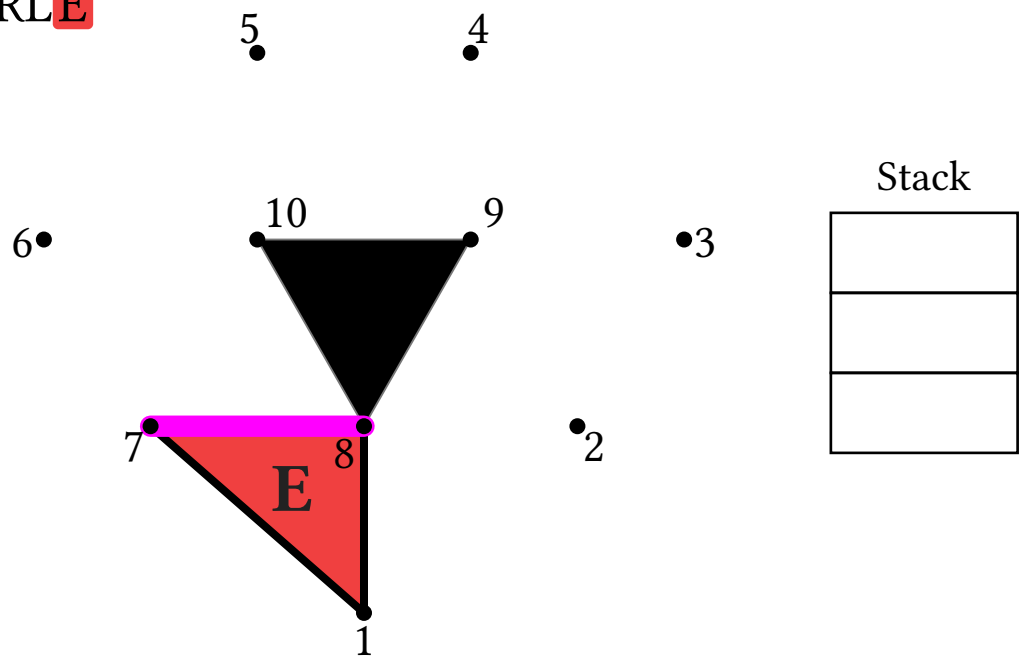


Figure 3: Komprese díry.

2.3 Vyšší genus?

History = SLRRRLRRLE

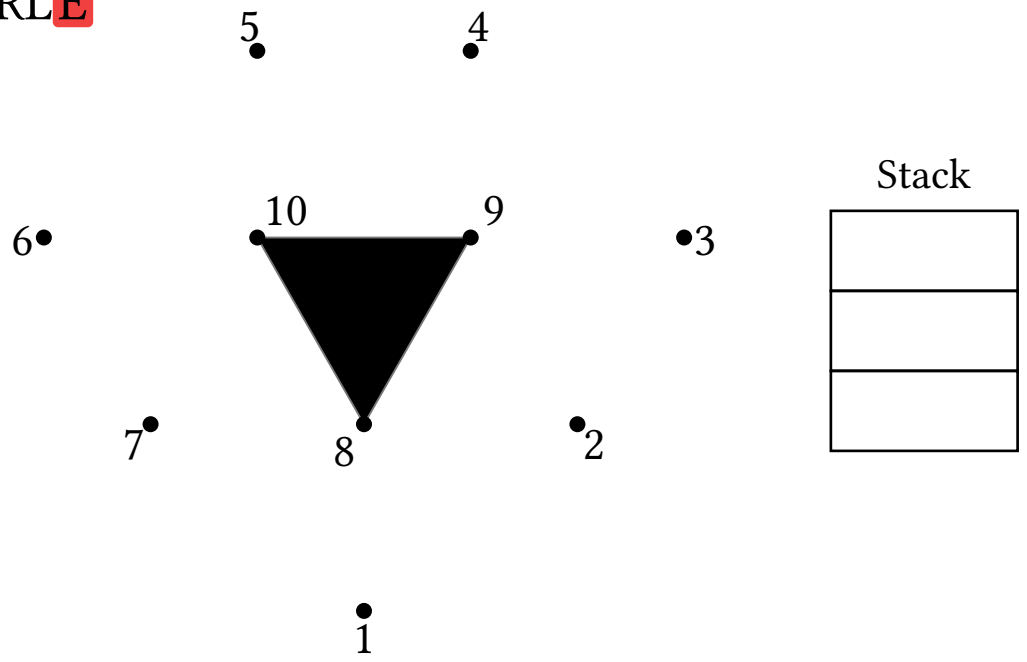


Figure 3: Komprese díry.

3. Dekomprese

3.1 Možnosti

3.2 Pravidla

History = CCRRRSLCRSERRELCRRRCRRRE

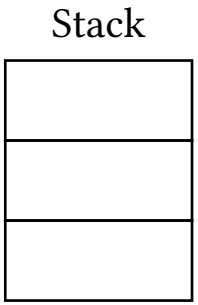


Figure 4: Dekomprese.

History = CCRRRSLCRSERRELCRRRCRRRE

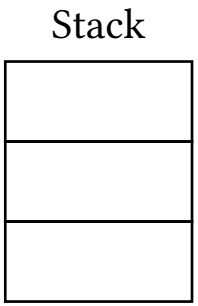


Figure 4: Dekomprese.

History = CCRRRSLCRSERRELCRRRCRRRE



Figure 4: Dekomprese.

History = CCRRRSLCRSERRELCRRRCRRR



Figure 4: Dekomprese.

4. Zdroje

4.1 Obrázky

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